

Child

1. Child depression **Fluoxetine**

Focus	Action	Responsibility
Detection	Risk profiling	Tier 1
Recognition	Identification in presenting children or young people	Tiers 2 to 4
Mild depression (including dysthymia)	Watchful waiting	Tier 1
	Psychological intervention	Tier 1 or 2
Moderate to severe depression	Child (aged 5–11 years) Psychological intervention with or without fluoxetine	Tier 2 or 3
	Young person (aged 12–18 years) Psychological intervention with or without fluoxetine	
Depression unresponsive to treatment/recurrent depression/psychotic depression	Intensive psychological intervention with or without fluoxetine, sertraline, citalopram, augmentation with an antipsychotic	Tier 3 or 4
June 2019 – the tiers terminology is under revision and may change in the future. NICE have retained the tiers terminology and will revise this when updated.		
Information from: NICE, 2019		

2. TTT of repetitive movements 12 months

- A. **Clonidine**
B. Haloperidol

[Psych-Mentor] A 12 year old boy presents to clinic with a 12 month history of vocal and motor tics. Further assessment reveals that he also has ADHD. Which of the following medication would be most appropriate to prescribe?

- A. **Clonidine**
B. Haloperidol
C. Risperidone
D. Atomoxetine
E. Pemoline

[Psych-Mentor] A child with ADHD using methylphenidate attends clinic. It becomes apparent during the consultation that they are suffering with tics. Which of the following alternative medications should you consider at this point?

- A. Dexamphetamine
B. **Atomoxetine**
C. Clonidine
D. Bupropion

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E. Modafinil

- ☞ Atomoxetine is recommended for ADHD where tics become a complication of stimulant use.
- ☞ Clonidine or Guanfacine considered in ADHD if there is **co-morbid tic diagnosis** (tic is not secondary to or side effect of stimulant)

3. boy agitated **behavior analysis**

4. male to female Tourette

A. 3:1

B. 1:1

Disorders with higher male preponderance		
Disease	Ratio	Higher in
Lewy body dementia	1.2:1	Males
Paranoid personality disorder	1.3:1	Males
Schizophrenia (Saha et al PLoSMed 2004) incidence rates only. Prevalence is equal.	1.4:1	Males
Parkinson's disease	1.5:1	Males
Conduct Disorder		Males
ADHD	3:1	Males
Alcohol dependence	5:1	Males
Korsakoff's psychosis	2:1	Males
Autism	3:1	Males
Asperger's syndrome Wing 1981 = 10:1	4:1	Males
Antisocial personality disorder	5:1	Males
Tourette's syndrome	4:1	Males

Condition	Sex ratio M:F
Reading disorder	3-4:1
Autistic spectrum disorder	4-5:1
Asperger's	5:1
Tourette's syndrome	2-5:1
ADHD (children)	2:1
ADHD (adults)	1.6-1
Major depression	1:2
Bipolar type 1	1:1
Panic with agoraphobia	1:3
Panic without agoraphobia	1:2
Generalised anxiety disorder	1:2
Obsessive compulsive disorder	1:1
Specific phobia	1:2
Conversion disorder	1:2-10
Anorexia nervosa	1:9
Bulimia nervosa	1:9
Borderline personality disorder	1:3

5. ICD11,DSM5 ADHD must have symptoms before the age of :

A. 12

B. 7

[ICD11] Attention deficit hyperactivity disorder is characterised by a persistent pattern (**at least 6 months**) of inattention and/or hyperactivity-impulsivity that has a direct negative impact on academic, occupational, or social functioning. **There is evidence of significant inattention and/or hyperactivity-impulsivity symptoms prior to age 12**, typically by early to mid-childhood, though some individuals may first come to clinical attention later.

OCD

6. OCD in child first line no sertraline in options

A. Fluvoxamine

B. Paroxetine

C. Citalopram

D. Escitalopram

E. Clomipramine

[SPMM] In the UK, which of the following drugs should be prescribed first in the treatment of obsessive-compulsive disorder in childhood?

Select one:

A. Fluoxetine

B. Clomipramine

C. Sertraline

D. Paroxetine

E. Fluvoxamine

Explanation: Sertraline (from age 6 years) and fluvoxamine (from age 8 years) are the selective serotonin reuptake inhibitors (SSRIs) licensed in the UK for the treatment of OCD in young people. Studies spanning 20 years have established the efficacy of SSRIs in the pediatric population in placebo-controlled trials. Sertraline and fluvoxamine are equally effective, but fluvoxamine may be less so. Paroxetine is not recommended for use in children and young people. On occasion, medications other than sertraline and fluvoxamine may be used as 'off-label' preparations with suitable caution. NICE guidance for the treatment of OCD recommends the use of SSRIs before the use of clomipramine, because of clomipramine's propensity for side effects and need for cardiac monitoring.

[Psych-Mentor] A 16 year old girl is brought to clinic by her mother. She is suffering with both obsessive compulsive disorder and depression. She has found psychological approaches unhelpful and wants to try medication. Which of the following is the most appropriate medication to try?

A. Fluoxetine

B. Clomipramine

C. Sertraline

D. Citalopram

E. Fluvoxamine

This scenario presents a dilemma, fluoxetine is the only antidepressant licensed for depression in people under 18, but only fluvoxamine and sertraline are licensed for OCD.

7. Paraphilia Compulsory sexual preoccupied OCD first line:

- A. **SSRI**,
- B. Clomipramine
- C. Aripiprazole

Treatment modalities for the paraphilias have limited scientific evidence to support their use. Psychological therapy (especially CBT) is often used (with extremely variable results).

Drug / class	Comments
SSRI	Data from small studies has shown some benefit in reduction of thoughts, urges, and behaviour
Naltrexone	Data from small study showed some reduction of thoughts, urges, and behaviour
Antipsychotics	Small studies have shown only weak treatment effects
GnRH agonists	Reduce testosterone. Have been shown to reduce paedophilic sexual thoughts/urges/desire and masturbatory frequency. Bone demineralisation along with hot flushes are potential side-effects. Relapse after stopping is consistently reported across different studies
Anti-androgens and progestational drugs (e.g. cyproterone acetate)	Reduce testosterone. Has been shown to reduce sexual outlet scores, spontaneous daytime erection and/or orgasm during masturbation in sex offenders. That said, responses are highly variable

8. Refractory OCD

- A. Clomipramine + Citalopram**
- B. Citalopram + prazosin

Box 3.7 Obsessive compulsive disorder	
First-line drug treatment (In order of preference)	
Drug	Comments
Any SSRI ²⁹ (Up to maximum licensed dose)	If the first SSRI is not tolerated or has a poor response an alternative SSRI may be tried ²⁹
Clomipramine (Up to 250mg)	Owing to poorer tolerability, recommended trying at least one SSRI first ²⁹
Second-line drug treatment (Unlicensed or weaker evidence base)	
Add antipsychotic to SSRI (Low to moderate doses of antipsychotics used in studies) ^{100,101}	Most evidence supports the use of aripiprazole or risperidone. ¹⁰⁰ Some evidence for haloperidol ¹⁰¹
Citalopram 40mg with clomipramine 150mg	Based on small randomised open label study ¹⁰² Recommended by NICE ²⁹ ECG monitoring required
Acetylcysteine ¹⁰³ up to 2400mg/day added to SSRI or clomipramine	Gastrointestinal adverse effects may be problematic. Two of five controlled studies negative. Pooled effect shows benefit ¹⁰⁴
Lamotrigine 100mg + added to SSRI ¹⁰⁵	Lamotrigine dose must be titrated gradually as indicated in the SPC May worsen OCD in some ¹⁰⁶
Topiramate up to 400mg added to SSRI ^{107,108}	Topiramate is not well tolerated, suggested benefits for compulsion but not obsessions. ¹⁰⁷ Two trials found topiramate ineffective ^{109,110}

9. NICE guideline OCD

A. Anterior Cingulotomy

B. rTMS (repetitive transcranial magnetic stimulation)

C. DBS

[SPMM] Ablative neurosurgery is perhaps more successful in OCD than in any other psychiatric disorder but is only employed in severe, treatment-resistant cases. Surgical treatment employs stereotactic methods using preoperative MRI to establish target coordinates. **The relative efficacy of both cingulotomy and ventral capsulotomy** (affecting white matter connections between the basal ganglia and cortex) in OCD is consistent with cortico-striatal involvement. Adverse events include intracranial hemorrhage, persisting postoperative side effects, cognitive and personality changes, and suicide.

Deep brain stimulation (DBS) is a new physical intervention in psychiatry that is increasingly being used outside of research settings. Which of the following is an indication for DBS?

Select one:

- ☐ Schizophrenia
- ☐ Body dysmorphic disorder
- ☒ OCD ✓
- ☐ Anorexia nervosa
- ☐ Bipolar affective disorder

Your answer is correct.

DBS is indicated in Parkinson's disease, which is refractory to or complicated by medical therapy. It can also be used in movement disorders (dystonia or tremors), depression, OCD, Tourette's Syndrome, epilepsy, cluster headache and chronic pain.

The correct answer is: OCD

Neurosurgery in OCD:

- ➔ **Anterior Cingulotomy → has been used for OCD → If not in the options then choose DBS**
- ➔ **rTMS has small evidence and more studies required, has been used in UK for treatment of depression but not OCD**

Evidence on the safety and efficacy of deep brain stimulation for chronic, severe, treatment-resistant obsessive-compulsive disorder (OCD) in adults is inadequate in quality and quantity. Therefore, this procedure should only be used in the context of research

10. DBS effective other than OCD: **Tourette, dystonia**

Explanation: DBS had shown to have therapeutic potential in patients with OCD. As well as OCD, there is evidence of therapeutic effects for **treatment resistant major depressive disorder, Parkinson's Disease among others.**

DBS is indicated in **Parkinson's disease**, which is refractory to or complicated by medical therapy. It can also be used in **movement disorders (dystonia or tremors), depression, OCD, Tourette's Syndrome, epilepsy, cluster headache and chronic pain.**

Addiction

11. Vivid dream, increase appetite, dysphoria:

A. Cocaine withdrawal

B. Amphetamine intoxication,

C. Alcohol withdrawal

Drug	UK Street names	Effects	Withdrawal features
Opioids	Heroin = Smack, horse, junk, skag, brown, Harry, H	Euphoria, drowsiness, constipation, pupillary constriction, respiratory depression	Piloerection, insomnia, restlessness, dilated pupils, yawning, sweating, abdominal cramps
Amphetamine and cocaine	Amphetamines = Speed, meth, ice, uppers, whizz. Cocaine = Snow, crack, coke, rock	Increased energy, insomnia, hyperactivity, euphoria, paranoia, tachycardia and hypertension reduced appetite, (for cocaine: formication, chest pain)	Hypersomnia, hyperphagia, depression(cocaine crash), irritability, agitation, vivid dreams, increased appetite
MDMA (ecstasy)	MDMA = XTC, E, Eccies, pills, Adam, Mandy	Increased energy, increased sweating, excessive water drinking and water intoxication lead to hyponatremia and death, jaw clenching , euphoria, enhanced sociability, increased response to touch	Depression, insomnia, depersonalisation, derealisation
Cannabis	Cannabis = Marijuana, weed, pot, grass, joint, mull, cone, ganja, hash, chronic	Relaxation, intensified sensory experience, paranoia, anxiety, injected conjunctivae	Insomnia, reduced appetite, irritability
Hallucinogens	LSD = Trips, Tripper, Tab, Stars, Smilies, Rainbows, Paper	Perceptual changes, pupillary dilation, tachycardia, sweating, palpitations, tremors, inco-ordination	No recognised withdrawal syndrome
Ketamine	Ketamine = Vitamin K, Super K, Special K, K, donkey dust	Euphoria, dissociation, ataxia, hallucinations, muscle rigidity	No recognised withdrawal syndrome

12. psychosis in withdrawal of cannabis, cocaine, heroin

Psychosis can occur in alcohol and benzodiazepine withdrawal

Syndromes of substance use

Match each clinical presentation with the most appropriate intoxication or withdrawal syndrome from the options listed.

1. A 35-year-old man presents with high anxiety, nightmares, profuse sweating, and insatiable hunger. He admits to feeling 'low' recently and thinking about hanging himself. ✖

2. A 19-year-old woman describes feeling very well (despite her family's concerns that she seems unlike herself), is slurring her speech, and is somewhat unsteady. She can't stop giggling and repeatedly asks if she can have something to eat. ✖

3. A 55-year-old man is found on the street to be unresponsive and barely breathing. It is noted that his pupils are constricted. ✖

4. A 24-year-old woman presents with hallucinations and a feeling of being 'outside' herself, though she doesn't seem particularly concerned. On examination, she is noted to be tachycardic and hypertensive, with bidirectional nystagmus. ✖

Heroin intoxication
Alcohol withdrawal
Ketamine intoxication
Cannabis intoxication
Cocaine withdrawal
GHB intoxication
Nicotine withdrawal

Explanation:

Cocaine withdrawal is characterised by a 'crash' with symptoms of anxiety, tremulousness, dysphoric mood, lethargy, fatigue, nightmares (accompanied by rebound REM sleep), headache, profuse sweating, muscle cramps, stomach cramps, and insatiable hunger. The most serious withdrawal symptom is depression, which can be particularly severe and associated with suicidal ideation or behaviour.

Effects of cannabis intoxication include mild euphoria ('the giggles'), a sense of enhanced well-being, a subjective sense of enhanced sensation, relaxation, altered time sense, and increased appetite ('the munchies'). Physically, there is mild tachycardia and variable dysarthria and ataxia.

Acute medical problems associated with heroin intoxication include nausea and vomiting, constipation, respiratory depression, and loss of consciousness with aspiration (the cause of many fatalities). Physical effects also include pupillary constriction and changes in blood pressure, heart rate, and body temperature.

Ketamine can cause hallucinations and a dissociated state in which the patient has an altered sense of the body and reality and little concern for the environment. On physical examination, the patient may be hypertensive and tachycardic, and have increased salivation and bidirectional or rotatory nystagmus.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 297, 305, 320-321. Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019. pp. 618, 626.

The correct answer is: (1) Cocaine withdrawal; (2) Cannabis intoxication; (3) Heroin intoxication; (4) Ketamine intoxication

13. Least dependence **LSD**

[SPMM] Among the following, the drug least likely to cause physical or psychological dependence is Select one:

- A. Cocaine
- B. Cannabis
- C. LSD**
- D. Amphetamine
- E. Alcohol

Explanation: LSD carries no risk of overdose, and physiological dependence and withdrawals do not occur.

14. Tactile sensation jaw rigidity amphetamine, ketamine

[Psych-Mentor] A man who has recently become unemployed has started to abuse drugs. His friend brings him to A&E as he started behaving strangely and appeared to be experiencing hallucinations. As he walks into the cubicle you note a very unsteady gait and further questions reveal a sense of muscle rigidity. Which of the following do you expect he has been using?

- A. Ketamine**
- B. LSD
- C. Cocaine
- D. Cannabis
- E. Diazepam

☞ **MDMA (Ecstasy) intoxication increases response to touch**

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[SPMM] Paul has acute onset auditory and visual hallucinations despite clear consciousness. His heart rate and temperature are elevated, and he is quite irritable. His presentation is most likely to be caused by Select one:

- A. Alcohol withdrawal
- B. Atropine poisoning
- C. Mushroom poisoning
- D. Amphetamine abuse**
- E. Delirium

Explanation: Acute harmful effects of amphetamines include tachycardia, arrhythmias, hyperpyrexia, irritability, post-use depression, and a quasipsychotic state with visual, auditory, and **tactile hallucinations**

[SPMM] The hallmark of amphetamine-induced psychosis is Select one:

- A. Inappropriate affect
- B. Predominance of tactile hallucinations.
- C. Paranoia**
- D. Predominant negative symptoms
- E. Formal thought disturbance

The hallmark of amphetamine-induced psychosis is the paranoia. It is very similar to schizophrenia but for the following; No negative symptoms, a predominance of visual hallucinations, associated **hyperactivity**, disinhibited sexual behaviour, confusion and incoherence and almost no formal thought disturbance

15. Long term cognitive impairment.

- A. Cannabis,
- B. **Cocaine**

[SPMM] 40-year-old man is a long term cocaine user. He is likely to have abnormalities in which of the following cognitive domains?

Select one:

- A. Sustained Attention**
- B. Alertness and arousal
- C. Abstract Reasoning
- D. Executive Function
- E. Verbal Fluency

Being a stimulant, cocaine enhances cognitive functions, especially response inhibition and psychomotor processing speed, acutely after administration, though long-term use impairs cognition. 30% of dependent users and even 12% of recreational users exhibit clinically relevant cognitive impairment (see Vonmoos et al., 2014). **The most impaired domains were sustained attention, response inhibition (impulsivity), verbal learning/memory, and working memory.** Abstinence helps to attain a level of cognitive performance level similar to non-users. Such recovery of cognition is not complete, as working memory impairments may continue despite 1-year of abstinence, especially in those who start cocaine use at an early age.

16. Mephedrone **stimulant**

[SPMM] Which of the following is the chemical form of Mephedrone?

Select one:

- A. Synthetic form of cocaine
- B. Synthetic drug of the cathinone**
- C. Synthetic form of steroids
- D. Synthetic form of cannabinoid
- E. Synthetic form of opioid

Explanation: Mephedrone is also known as 4-methylmethcathinone (4-MMC) or 4-methylephedrone. It is a **synthetic stimulant drug** of the amphetamine and also the cathinone compounds found in the khat plant of Eastern Africa. It produces similar effects to cocaine, MDMA and amphetamines.

known as meow-meow, is a white crystal or a powdery substance, sometimes sold as capsules or pills. It has a **distinctive fishy odour** though some products are described to have vanilla, bleach, or stale urine smell. This smell is often notable in the sweat of the user.

17. Best predictor antecedent behaviour of gaming addiction in young people:

- A. Introvert
- B. Novelty seeking behaviour,**
- C. Less neurotism introversion,
- D. Rich fantasy life

Individuals addicted to videogames need novelty seeking, socialization, competition, and/or entertainment.

Gaming disorder (ICD-11)⁸

Classified under 'Disorders due to addictive behaviours' in ICD-11, together with gambling disorder, gaming disorder is characterized by a pattern of persistent or recurrent gaming behaviour ('digital gaming' or 'video-gaming'), which may be online (i.e. over the Internet) or offline. Current epidemiological studies estimate prevalence as ~1–27%, but there are marked differences in diagnostic methods used and populations studied—more research is definitely needed.

Diagnostic criteria

- Impaired control over gaming (e.g. onset, frequency, intensity, duration, termination, context);
- Increasing priority given to gaming, to the extent that gaming takes precedence over other life interests and daily activities; and
- Continuation or escalation of gaming despite the occurrence of negative consequences.
- The behaviour pattern is of sufficient severity to result in significant impairment in personal, family, social, educational, occupational, or other important areas of functioning.
- The pattern of gaming behaviour may be continuous or episodic and recurrent. The gaming behaviour and other features are normally evident over a period of at least 12 months in order for a diagnosis to be assigned, although the required duration may be shortened if all diagnostic requirements are met and symptoms are severe.

Comorbidity Depression, ADHD, alcohol misuse, anxiety, and lack of psychosocial supports.

Treatment Address any comorbid disorder. Counselling and CBT/behavioural modification are key to specific interventions. Some evidence for use of bupropion. Self-help (12-Step Programme) such as through On-line Gamers Anonymous (J&S <http://www.olganon.org/home>).

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9831379/>

18. Detoxification inpatient when ? History taking BNZ

Question about taking detailed history first before starting detoxification. ??

19. Chronic Alcohol imaging

A. Corps callosum enlargement,

B. Lateral ventricle,

C. Whit matter anisotropy

[SPMM] Which of the following types of alcohol-related neurological damage has an association with red wine?

Select one:

A. Alcoholic dementia

B. Hepatocerebral degeneration

C. Alcoholic hepatic encephalopathy

D. Cerebellar ataxia

E. Marchiafava-Bignami syndrome

Explanation: CNS myelin is vulnerable to numerous toxins. For example, a contaminant of certain homemade Italian red wines probably causes degeneration of the heavily myelinated corpus callosum, causing the Marchiafava-Bignami syndrome. Theoretically, at least, this syndrome can damage the corpus callosum enough to produce a 'split brain syndrome'. Patients show disconnection signs and extensive cerebral dysfunction.

[Psych-Mentor] Which of the following is characterised by progressive demyelination of the corpus callosum and is generally associated with alcoholism?

A. Hunter's syndrome

B. Hurler's syndrome

C. Schilder's disease

D. Wilson's disease

E. Marchiafava-Bignami disease

20. Antecedent stimulus control

Google: An antecedent is a stimulus that cues an organism to perform a learned behavior. When an organism perceives an antecedent stimulus, it behaves in a way that maximizes reinforcing consequences and minimizes punishing consequences. This might be part of complex, interpersonal communication.

ABC: Antecedent – Behaviour and Consequences

Behavioral approaches may look at antecedents, behaviour and consequence charts (ABC charts) and are helpful in the management of problem behaviour.

LD

21. Noisy, hypertelorism, microcephaly, hypotonia, moderate ID **Cri du chat**

[SPMM] A 5-year-old boy with severe intellectual disability has a round face with hypertelorism, slanting palpebral fissures, broad flat nose, and an infantile cat-like cry. Which of the following conditions is he most likely to suffer from?

Select one:

A. Cri-du-chat syndrome

B. Down syndrome

- C. Angelman syndrome
- D. Smith-Magenis syndrome
- E. Williams syndrome

Explanation: Cri-du-chat syndrome is a partial monosomy; karyotype 5p- (varies from deletion of a small band at 5p15.2 to the entire arm of 5p). The infant has a characteristic **high-pitched cry** that resembles a cat. Clinical features include **microcephaly, rounded face, hypertelorism, micrognathia**, dental malocclusion, epicanthic folds, low-set ears, **hypotonia**, and severe/profound intellectual disability.

Hypertelorism refers to an abnormally increased distance between two organs or bodily parts, usually referring to an increased distance between the orbits (eyes). It is seen in some conditions like Di George's syndrome, Noonan syndrome, Edwards syndrome, cri du-chat syndrome and neurofibromatosis. Salient features of DiGeorge syndrome can be summarized using the mnemonic CATCH-22, with the 22 reminding us of chromosome 22, and CATCH referring to Cardiac Abnormality, Abnormal facies (palatal abnormalities, hypertelorism), Thymic aplasia, Cleft palate, Hypocalcaemia / Hypoparathyroidism.

Learning Disability Syndromes	
Which is the most likely diagnosis for each of the given patients that have been referred to your LD clinic?	
A 5-year-old boy with short stature and hypertelorism . His mother complains that he is hyperactive and very noisy.	Choose... ☐ Klinefelters syndrome ☐ Smith Magenis Syndrome ☐ Edward's syndrome ☐ Cri-du-chat syndrome ☐ Prader Willi Syndrome ☐ Down's Syndrome ☐ Fragile X Syndrome ☐ Turner's Syndrome ☐ Landau Kleffner Syndrome Choose...
An 8-year-old boy who suffers from seizures and has recently developed mutism	Choose...
A short 12-year-old girl with a webbed neck, whose recent CT scan revealed a narrowed aorta	Choose...
A 14-year-old boy who has a history of social anxiety. In clinic he appears shy and avoids looking at your directly.	Choose...
A 9-year-old obese girl has a history of obsessional behaviour involving food that includes hyperphagia. She has a borderline IQ.	Choose...

Explanation:

- 1) **Cri-du-chat syndrome**, also known as chromosome 5 depletion is a rare genetic disorder characterized by inappropriate laughter and a cat-like cry during infancy
- 2) **Landau Kleffner Syndrome** is an epilepsy syndrome of childhood. Its onset usually occurs before the age of six and is twice as likely to affect boys than girls.
- 3) **Turner's Syndrome** is a genetic disorder that only affects females causing distinctive characteristics such as short stature, webbed neck, widely spaced nipples and coarctation of the aorta.
- 4) **Fragile X syndrome** is an inherited cause of intellectual disability resulting in characteristics such as social anxiety, gaze avoidance and shyness.
- 5) **Prader-Willi syndrome** is a rare genetic disorder characterized by mild-to-moderate learning disability, food obsession, insatiable appetite, hyperphagia and obesity. Male : female 1:1

22. Fragile X most common symptoms **hyperactivity**

Behavioural features include **hyperactivity**, inattention, anxiety, stereotypies, speech and language delays, IQ decline, repetitive mannerisms, shyness, gaze avoidance, poor peer relationships, and communication difficulties.

[SPMM] Which of the following is a common feature of fragile-x syndrome?

Select one:

- A. Catatonia
- B. Dissociative episodes
- C. Compulsive hand-washing
- D. Cluttering speech**
- E. Recurrent fractures

Fragile X speech is often described as "cluttered". Poor topic maintenance with frequent tangential comments may occur. **Syntax (arrangement of words and phrases)** is usually **appropriate for mental age**; a high receptive vocabulary score is usually seen, and auditory sensory and processing skills are weak.

23. MECP2 **Retts**

[SPMM] The MECP2 gene and protein appear to be essential for the normal function of nerve cells. Abnormalities within the gene are seen in which of the following disorders?

Select one:

- A. Angelman syndrome
- B. ADHD
- C. Fragile X syndrome
- D. Rett Syndrome**
- E. ASD

Rett syndrome is mainly found in **girls** with a prevalence of around **1/10,000**. Around 6-18 months after birth, speech and motor function capabilities start to decrease. This is followed by seizures, growth retardation and cognitive and motor impairment. The MECP2 locus is X-linked and the disease-causing alleles are dominant.

[SPMM] Jenny is a 3-year-old girl with global developmental delay. She did learn to speak a few words initially but has lost this ability now. She demonstrates breathing difficulties, odd hand movements and also **suffers seizures**. Which of the following disorders cause epilepsy in childhood?

Select one:

- A. Boerhaave syndrome
- B. Down Syndrome
- C. Rett Syndrome**
- D. Gilbert Syndrome
- E. DiGeorge (22q11.2 deletion) syndrome

Rett syndrome is a neurodevelopmental disorder caused by mutations in **MECP2** that occurs almost exclusively in females. After a period of initially normal development during the first 6 to 18 months of life, girls experience loss of speech and purposeful hand use, and develop stereotypic hand movements, and gait abnormalities. Epilepsy affects the majority of patients with the syndrome. DiGeorge syndrome

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affects the palate and heart. Gilbert Syndrome causes benign raised bilirubin levels. Boerhaave syndrome occurs when vomiting causes esophageal tears. Down Syndrome affects a number of organ systems but is not associated with seizures.

24. Case Child 11 y Epilepsy, language 6 years

A. Landau kleffener

B. Retts

C. Corneil

See Q.21

25. Equal distribution **pardal willi**

Prader-Willi syndrome

- * **Prader-Willi** syndrome (PWS) is a rare genetic disorder associated with deletion or inexpression of 7 genes on chromosome 15 (q11-13) on the paternal chromosome. Males are more commonly affected, M:F = 4:3. Incidence of between 1/10,000 and 1/25,000 live births.
- * Early characteristic features:
 - o Hypotonia
 - o Lethargy
 - o Hypogonadism
 - o Often breech or caesarean births
 - o Failure to thrive in infancy
 - o Excessive sleepiness
 - o Triangular mouth causing feeding and swallowing problems in infancy
- * In early childhood, there is a marked increase in the frequency of over-eating; 80% of affected individuals have hyperphagia. Affected individuals have an insatiable appetite leading to stealing of food and can sometimes lead to the consumption of unpalatable food.

➡ **Child OCD and Bipolar have equal distribution as well**

26. LD aggression **behavior analysis**.

[SPMM] A person with intellectual disability has aggressive behavioural issues. Which of the following is the best first step in approaching this situation?

Select one:

A. Functional behaviour assessment/analysis

B. Behaviour support plan

C. Prescription of an antipsychotic

D. Psychoeducation

E. Physical restraints

Behaviour disorders are over-represented in intellectual disability (ID) populations. It is important to perform a thorough assessment, including functional assessment of the behaviour (i.e. description of the behaviour, situations, consequences, and reinforcers) before implementing management strategies. Originating in the mid-1960's, functional behavioural analysis has become the gold standard for understanding the environmental variables that come to shape and maintain problematic behaviours such as aggression, self-injury, and property destruction.

27. LD menstruation abd pain Intellectual disability girl getting angry during her period and hitting her stomach

- A. Levonorgestrel
- B. Ibuprofen**
- C. SSRI

28. LD pt moderate ID refuse injection withdraw blood for thyroid test:

- A. Flooding,
- B. Desensitization,
- C. Distraction,
- D. Give bnzo
- E. Risperidone

It depends on the options, we are between Benzodiazepines and distraction. Desensitization might take time and be difficult in moderate LD.

29. LD release of individuals from institutional care such as a psychiatric hospital to care in the community

Normalization

[SPMM] Mr Jason Hughes is a 42-year-old man with a diagnosis of Down's Syndrome and mild intellectual disability. He largely lives independently with some community support and works for a local charity. He and his family feel it is important that he has a social role and life circumstances that are not too different from others of his age. Which of the following best describes this social phenomenon?

Select one:

- A. Deinstitutionalisation
- B. Assimilation
- C. Normalisation**
- D. Acclimatisation
- E. Social role valorisation

Normalisation refers to the principle that people with intellectual disabilities should be able to experience 'normal patterns' of everyday life, such as places of accommodation, and the range of day-to-day activities. Rather than forcing individuals to conform to societal norms, the principle argues for changing the environment, not the individual, to provide the required normalcy.

Social role valorization emphasizes the importance of enabling and supporting individuals to occupy valued social roles within their communities. The theory suggests that individuals who hold valued social roles are more likely to experience social inclusion, acceptance, and opportunities for growth and development.

Key principles of social role valorization include:

1. Valued social roles: Individuals are more likely to experience social acceptance and inclusion when they occupy roles that are valued and respected within their communities, such as parent, friend, employee, or volunteer.
2. Normalization: This principle advocates for ensuring that individuals with disabilities have access to the same opportunities, experiences, and lifestyles as their non-disabled peers to the greatest extent possible. It promotes integration and inclusion rather than segregation or isolation.

3. Enhancement of personal competencies: Social role valorization emphasizes the importance of supporting individuals to develop and enhance their personal competencies, skills, and abilities. This may involve providing education, training, and opportunities for skill development to help individuals succeed in their chosen roles.
4. Social support networks: Building and maintaining social support networks is crucial for individuals to thrive in their valued social roles. This may involve fostering connections with family, friends, peers, and community members who provide encouragement, guidance, and assistance as needed.

Overall, social role valorization encourages society to recognize and value the contributions of all individuals, regardless of their abilities or disabilities, and to create environments that support their participation and inclusion in meaningful social roles.

Psychotherapy

30. DPT correct statement:

- A. It comprises of a hierarchy of goals,**
- B. It requires group participation,
- C. It includes goodbye letters,
- D. Low frustration tolerance is a contraindication,
- E. It is based on psychodynamic formulations

[SPMM] Which of the following is correct about dialectical behaviour therapy?

Select one:

- A. It is only done in groups
- B. It is built on a hierarchy of goals**
- C. It includes goodbye letters
- D. Low frustration tolerance is a contraindication
- E. It is based on psychodynamic formulations

Explanation: Dialectical behaviour therapy (DBT) was introduced in 1991 by Linehan and colleagues as a treatment for borderline personality disorder (BPD). Patients with BPD are supported in understanding their own emotional experiences and are taught new skills for dealing with their distress through a combination of group and individual therapy sessions. DBT is a complex treatment combining CBT interventions with Eastern meditative practice, notably mindfulness. **DBT takes a hierarchical view of treatment aspirations, with the focus first on reducing behaviours that cause self-harm, then on reducing those behaviours that interfere with therapy, and finally aiming to reduce behaviours that diminish the quality of life and personal relationships**

31. PTSD children

- A. Ask child direct**
- B. Avoid allow child initiate.
- C. Ask family.

[SPMM] An 11-year-old boy presents with possible symptoms of PTSD. How should his case be handled?

Select one:

- A. Ask the child directly about his experiences**
- B. Rely on the caregivers to give the history
- C. Offer immediate debriefing
- D. Do not ask direct questions; allow the child to initiate
- E. Speak with the family about drug treatments

Identification of PTSD in children can be improved by asking the child directly about their experiences. Do not rely solely on the caregivers' history. Drug treatments should not be routinely prescribed for children and young people with PTSD.

32. CBT depression, he is struggling to motivate himself to engage with his usual activities and is due to be assessed for behavioural activation therapy. Which of the following will this include?

A. Degree of avoidance,

- B. Negative cognitions,
- C. Impulsivity ,
- D. Logic errors,
- E. Early life activities

[SPMM] Mr Neil Peterson is a 36-year-old man who is currently experiencing a moderate depressive episode. He is struggling to motivate himself to engage with his usual activities and is due to be assessed for behavioural activation therapy. Which of the following will this include?

Select one:

A. Degree of avoidance

- B. Negative cognitions
- C. Logic errors
- D. Early life activities
- E. Impulsivity

Assessment for behavioural activation therapy includes **identification of TRAP (Trigger, Response, Avoidance Pattern)** with the avoidance including direct avoidance, rumination and use of alcohol, drugs and other maladaptive coping strategies. The aim is to replace this with TRAC (Trigger, Response, Alternate Coping response).

33. psychotherapy focus of social milieu transaction analysis scheme,

- A. CAT
- B. IPT
- C. Compassion focused therapy

[SPMM] Tom is a 12-year-old boy with severe depression. He was offered CBT that has failed. His parents are worried about side effects of medication and don't want him to have pharmacotherapy. The best alternative in this situation is

Select one:

- A. Parenting training
- B. Dialectical behavioural therapy
- C. Interpersonal therapy**
- D. Mindfulness training
- E. Cognitive analytical therapy

Interpersonal Therapy (IPT) and **Transactional Analysis focus in particular on interpersonal relationships**, and both may be effective in cases of depression. IPT has been adapted for use with children and adolescents. **The areas of focus in IPT include conflicts with others, losses, changes in circumstances/difficulties adapting to new circumstances, and social isolation.** Transactional Analysis looks at the way in which people take up roles of Parent, Adult or Child, exploring inner dialogue between different components of the self.

34. What is element of motivational interviewing?

A. Cognitive dissonance amplification,

B. Incentive,

C. Goal setting

[SPMM] Which of the following techniques aims to amplify cognitive dissonance?

Select one:

A. Motivational interviewing

B. Flooding/implosive therapy

C. Exposure and response prevention

D. Systematic desensitisation

E. Token economy

Cognitive dissonance theory is founded on the assumption that individuals seek consistency between their expectations and their reality and will strive to reduce the discomfort that any discrepancy between the two causes. Motivation for change builds when individuals start to perceive a discrepancy between current and desired behaviour (ambivalence) and then options to resolve the cognitive dissonance are identified. This is usually the first, (directive) step in motivational interviewing for eliciting behaviour changes, a technique with an established evidence base.

Motivational interviewing (MI) is a technique used to motivate the patient to change maladaptive behaviour. The therapist relies on empathy to convey understanding, provides support by noting the patient's strengths, and explores the ambivalence and conflicting thoughts or feelings that the patient may have about change. Principles of MI include expressing empathy (not sympathy), developing discrepancy, rolling with resistance, and supporting self-efficacy and optimism. Open questioning is one of the skills that is used to explore ambivalence and elicit change talk.

DEARS

D- Developing discrepancy

E- Empathy

A- Avoid arguments

R- Rolling with resistance

S- Support self- efficacy

35. Transference choose true statement:

A. **May be psychotic ,**

B. Related to therapist only,

C. Should be ignored ,

D. It takes months for it to happen ,

E. It is explained by mechanism of sublimation

[SPMM] Mr Carl Sullivan is a 38-year-old man who has a diagnosis of OCD. He was assessed and referred for CBT. He finds himself being attracted to the therapist and subsequently joins a gym. Which of the following is he demonstrating?

A. Rationalisation

B. Displacement

C. Repression

D. Sublimation

E. Regression

Explanation: This is most likely to be sublimation. The patient is experiencing an erotic transference towards the therapist and, rather than acting on it in an inappropriate way, directing it towards a more appropriate activity. This erotic attraction is a form of positive transference, whereby the therapist is viewed as helpful and a good influence but taken further. It is a relatively common phenomenon.

Adult

36. EGFR <5 DOC

A. Olanzapine

- B. Clozapine
- C. Chlorpromazine
- D. Sulpiride
- E. Amisulpride

[SPMM] An 80-year-old woman with an eGFR 25 is to be prescribed an antipsychotic. Which of the following drugs would be the most appropriate for her?
Select one:

A. Olanzapine

- B. Chlorpromazine
- C. Clozapine
- D. Amisulpride
- E. Sulpiride

In the setting of renal impairment, no antipsychotic is clearly preferred to another. However, Sulpiride and Amisulpride are extensively renally cleared and should be avoided. Avoid highly anticholinergic agents (such as clozapine and chlorpromazine) because they can contribute to urinary retention. **Recommended choices are haloperidol and olanzapine (5 mg in general adults, 2.5mg in elderly).**

Table 8.8 Antipsychotic medications in renal impairment, Maudsley Guidelines P. 738

Olanzapine ^{9,22,23,25,40,44}	57% of olanzapine is excreted mainly as metabolites (7% excreted unchanged) in urine. Dosing: GFR <50mL/min initially 5mg daily and titrate as necessary. Avoid long-acting preparations in renal impairment unless the oral dose is well tolerated and effective. Manufacturer recommends a lower long-acting injection starting dose of 150mg, 4-weekly in patients with renal impairment. May cause and aggravate diabetes, a common cause of renal disease. Hypothermia has been reported when used in renal failure.
--	---

37. **Memantine with caution eGFR 53/1.73m², epilepsy**

^{BNF} Reduce dose to 10 mg daily if eGFR 30–49 mL/minute/1.73 m², if well tolerated after at least 7 days dose can be increased in steps to 20 mg daily; reduce dose to 10 mg daily if eGFR 5–29 mL/minute/1.73 m².

Use with Caution: Cautions For memantine hydrochloride Epilepsy; history of convulsions; risk factors for epilepsy

If question asking using memantine with caution then memantine could be answer ??

38. Epilepsy with psychosis

- A. Haloperidol
- B. Quetiapine
- C. Risperidone
- D. Lurasidone

Antipsychotics		
Low risk – good choices	Amisulpride/sulpiride	Considered to be safe in PWE. ⁴³ Renally excreted, so low risk of pharmacokinetic interactions with antiseizure medications. Seizures uncommon in overdose ⁴⁴
	Aripiprazole	Rarely lowers seizure threshold. ⁵ Incidence of seizures similar to placebo in RCTs ²⁴
	Ziprasidone	
	High potency FGAs	For example, fluphenazine, haloperidol, trifluoperazine, flupentixol. Low risk of lowering the seizure threshold ⁵
	Risperidone	Unlikely to lower the seizure threshold. ⁵ Incidence of seizures similar to placebo in RCTs. ²⁴ Has been recommended for PWE. ^{32,45} Evidence of safety in a case series of adolescents with epilepsy ⁴⁶

[Click her for full table](#)

We believe question might be which to avoid and answer would be **Quetiapine**

39. Tamoxifen avoid **Paroxetine**

Mrs. Y is treated for breast cancer with Tamoxifen. She is clinically depressed. Which of the following drugs is contraindicated in her situation?

Select one:

- A. Amitriptyline
- B. Mirtazapine
- C. Fluoxetine
- D. Reboxetine
- E. Venlafaxine

In several studies (notably, Jin et al, 2005), concurrent use of tamoxifen with the potent CYP2D6-inhibitor antidepressants **paroxetine and fluoxetine**, was associated with a 65-75% reduction in circulating endoxifen (active metabolite of tamoxifen) levels in some women (especially in poor metabolisers). Based on current research into psychotropics, the **strongest** CYP2D6 inhibitors are identified as **paroxetine, fluoxetine, bupropion, duloxetine**, while **sertraline, escitalopram, and doxepin** are **moderate** inhibitors, with **venlafaxine** being a **weak inhibitor**. Therefore, if antidepressants are indicated in the treatment of a woman currently taking tamoxifen, avoid antidepressants such as fluoxetine and paroxetine, bupropion and duloxetine. Co-administration with other potent CYP2D6 inhibitors (e.g. quinidine, or bupropion) should also be avoided whenever possible.

BANPDF = **B**upropion, **A**mitriptyline (TCAs), **N**?**Paroxetine**, **D**uloxetine, **F**luoxetine?

A 46-year-old lady has been treated with Tamoxifen for breast cancer. She is also clinically depressed and will need treatment with antidepressants. Which one among the following should be avoided due to its effects upon Tamoxifen?

Select one:

- A. Mirtazapine
- B. Venlafaxine
- C. Amitriptyline
- D. Duloxetine
- E. Moclobemide

In several studies (notably, Jin et al, 2005), concurrent use of tamoxifen with the potent CYP2D6-inhibitor antidepressants such as duloxetine, bupropion, paroxetine and fluoxetine, was associated with a significant reduction in circulating endoxifen levels in some women and therefore should be avoided.

[SPMM] Which of the following drugs is best avoided in a patient taking Tamoxifen?

Select one:

- A. Fluoxetine**
- B. Sertraline
- C. Citalopram
- D. Venlafaxine
- E. Amitriptyline

Tamoxifen prevents recurrence and death from breast cancer. As a prodrug, tamoxifen requires conversion to active metabolites, the most important of which is endoxifen, in a process influenced by CYP2D6. SSRIs that inhibit CYP2D6, such as paroxetine and fluoxetine, are best avoided in patients taking tamoxifen. Sertraline, citalopram, escitalopram, and venlafaxine impart lesser degrees of inhibition and are reasonable alternatives.

Citalopram is a highly selective serotonin reuptake inhibitor metabolized by the cytochrome P450 2C19 enzyme and the cytochrome P450 3A4 enzyme. The cytochrome P450 2D6 enzyme also plays a secondary role in metabolizing citalopram

40. Risperidone 2mg, Citalopram 20mg QTC 490

- A. Reduce risperidone**
- B. Refer to cardiology immediately.

[SPMM] An elderly man taking sertraline 50mg OD and risperidone 2mg OD is found to have a QTc of 500ms. What is the most appropriate next step of action?

Select one:

- A. Refer to cardiology urgently**
- B. Switch risperidone to quetiapine
- C. Discontinue sertraline
- D. Repeat ECG in a week
- E. Decrease the dosage of risperidone

QTc	Action	Refer to cardiologist
<440ms (men) or <470ms (women)	None unless abnormal T-wave morphology	Consider if in doubt
>440ms (men) or >470ms (women) but <500ms	Consider reducing dose or switching to drug of lower effect; repeat ECG	Consider
>500ms	Repeat ECG. Stop suspected causative drug(s) and switch to drug of lower effect	Immediately
Abnormal T-wave morphology	Review treatment. Consider reducing dose or switching to drug of lower effect	Immediately

[SPMM] A 50-year-old man with a history of paranoid schizophrenia has been prescribed 600 mg of Quetiapine. During a routine screening with ECG, he has a QTc interval of 510 ms. He does not report any cardiac symptoms. The appropriate next step would be

Select one:

- A. Cross taper quetiapine with Olanzapine and repeat ECG in 2 weeks
- B. Refer to cardiology immediately
- C. Repeat ECG in 2 weeks, since patient is asymptomatic
- D. Repeat ECG and refer to cardiology if T wave is abnormal

E. Reduce the dose of Quetiapine, repeat ECG and consider referral to cardiology

Explanation: Strong evidence links QTc values over 500 ms to a clearly increased risk of arrhythmias. The dosage should be reduced immediately and consider referral to a cardiologist immediately.

[Psych-Mentor] A patient prescribed quetiapine (500mg once daily) had a recent ECG which showed a QTc interval of 510ms. Which of the following would be the most appropriate next step?

- A. Switch to haloperidol and then repeat ECG
- B. Reduce dose of quetiapine, repeat the ECG, and refer for a cardiology opinion
- C. Stop quetiapine completely and then switch to aripiprazole and refer to cardiology**
- D. Do nothing as long as the patient is asymptomatic
- E. Cross-taper in order to start risperidone and repeat ECG in 2 weeks

Exam Question Oct 2010: Haloperidol is thought to have a high effect on the QTc interval. In situations where the QTc interval is high, aripiprazole and olanzapine are best but risperidone would be a reasonable alternative.

41. psychogenic rather than epilepsy

- A. Moving head side to side,**
- B. Open eye,
- C. Always continence urine

[SPMM] Which of the following favours psychogenic nonepileptic seizures rather than epileptic seizures? Select one:

- A. Opened eyes during seizures
- B. Presence of automatisms
- C. Total duration less than 30 secs
- D. Predictable anatomical progression of seizures
- E. Side-to-side movements of the head**

Explanation: Ictal eye closure has been assessed in several controlled studies and is significantly more frequent in psychogenic nonepileptic seizures (PNES) than epileptic seizures (ES). Closed eyes during an attack is considered a discriminator between PNES and ES. The mean duration of PNES is longer than ES. There is considerable evidence to suggest that a duration of more than 2 minutes is highly suggestive of PNES, although this is an arbitrary limit. Duration of longer than 10 minutes strongly suggests PNES. In studies that have compared convulsive PNES with generalised tonic clonic seizures, the proportion of events with side-to-side head or body movement, especially with coordinated agonist and antagonist activity, are significantly higher in PNES.

Signs & symptoms favouring psychogenic nonepileptic seizures (PNES)	Signs & symptoms favouring epileptic seizures (ES)
Duration longer than 10 minutes	
Vocalisation during or after seizures, complex, with affective content	Vocalisation at the beginning, primitive, has no emotional expression
Ictal crying	
Occurrence from preictal pseudosleep	Occurrence from sleep
Ictal eye closure	
Side-to-side head or body movement (compared with generalised tonic clonic seizures only)	Postictal stertorous breathing (observed in generalised tonic clonic seizures only)
Pelvic thrusting (frontal lobe partial seizures excluded)	
Rapid regaining of alertness and orientation	Post ictal confusion
Fluctuating course	
Asynchronous jerks (frontal lobe partial seizures excluded)	
Memory recall	

Adapted from Xiang et al. 2019

42. Predictive long term cognition impairment

A. Duration of PTA ,

B. duration of unconsciousness,

C. GCS

[SPMM] Poor prognostic factors with respect to recovery following traumatic brain injury include all except

Select one:

A. Long period of post-traumatic amnesia

B. Long duration of loss of consciousness

C. Older age

D. Focal damage restricted to the non-dominant lobe

E. Chronic alcohol dependence

Explanation: Factors associated with poorer long-term outcome after traumatic brain injury include the conscious level post-injury; duration of loss of consciousness; increased duration of post-traumatic amnesia (>24 hours, poorer outcome); age (older has poorer prognosis); pre-injury educational or occupational level; and reduced pre-injury cognitive reserve (e.g. due to cerebrovascular disease or alcohol dependence).

Duration of PTA	Classification	Functional outcome
PTA less than 60 minutes	Mild injury	May return to work in <1 month
PTA between 1-24 hours	Moderate injury	May return to work in 2 months
PTA between 1-7 days	Severe injury	May return to work in 4months
PTA greater than 7 days	Very severe injury	May require > 1 year for return to work

43. Which one is a sequelae of head injury?

A. **ADHD,**

B. ASD,

C. Personality disorders

44. Bulimia other than anorexia **bulimia has higher prevalence than anorexia**

[SPMM] Which of the following suggests a diagnosis of bulimia nervosa rather than anorexia nervosa?

Select one:

A. Age of onset at 15

B. Lanugo hairs

C. Calorie counting

D. Laxative abuse

E. Poor impulse control

Anorexia	Bulimia
Onset mostly in adolescents	Mostly young adults; little later onset
Excess in higher social class	Even class distribution
0.5-1% prevalence in teenage girls	1-2% prevalence in 16-35 age group
19/100 000 females per year	29/100 000 females per year

(adapted from Fairburn & Harrison, 2003)

Explanation: There is an increased prevalence of disorders of impulsivity and lack of control in people with bulimia nervosa. Patients who have to satisfy their binges at all costs (e.g. shoplifting) may have issues with impulsivity and rage. People with bulimic illnesses also have a higher prevalence of substance misuse.

45. First line in treatment of Bulimia

- A. Guided self help,
- B. individual CBT ,
- C. fluoxetine

For adults, the first step is an **evidence-based self-help programme**.

If bulimia-nervosa-focused guided self-help is unacceptable, contraindicated, or ineffective after 4 weeks of treatment, **consider individual eating-disorder focused cognitive behavioural therapy (CBT-ED)**.

For **children and young people offer bulimia-nervosa-focused family therapy (FT-BN)**.

If FT-BN is unacceptable, contraindicated or ineffective, consider individual eating-disorder-focused cognitive behavioural therapy (CBT-ED) for children and young people with bulimia nervosa.

Do not offer medication as the sole treatment for bulimia nervosa.

[SPMM] Miss. X is a 16-year-old girl who was assessed in the CAMHS outpatient clinic and diagnosed with bulimia nervosa. Which of the following psychotherapeutic interventions is the most effective treatment in this age group?

Select one:

- A. Family therapy
- B. Behavioural therapy
- C. Psychodynamic psychotherapy
- D. CBT
- E. Interpersonal therapy

Explanation: Adolescents with bulimia nervosa may be treated with CBT adapted, as needed, to suit their age, circumstances, and level of development, and including the family as appropriate. NICE advises considering bulimia-nervosa-focused guided self-help (using cognitive behavioural materials) supplemented with brief supportive sessions. If this is unacceptable or ineffective after 4 weeks, consider individual CBT-ED for bulimia nervosa.

Bulimia nervosa

• Psychological treatments in adults:

- Bulimia-nervosa-focused guided self-help programmes may be considered for example 4 to 9 sessions lasting 20 minutes each over 16 weeks, running weekly initially.
- If bulimia-nervosa-focused guided self-help is unacceptable, contraindicated, or ineffective after 4 weeks of treatment, individual eating-disorder focused cognitive behavioural therapy (CBT-ED) may be considered. This typically consists of up to 20 sessions over 20 weeks.

• Psychological treatments in children and young people:

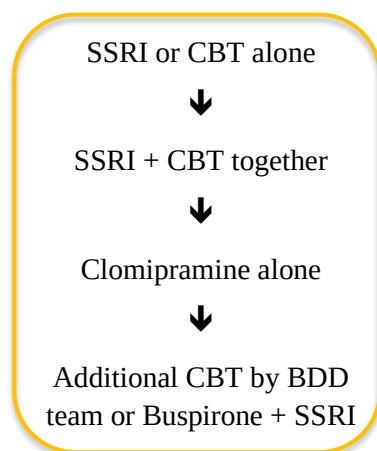
- Bulimia-nervosa-focused family therapy (FT-BN) may be considered in children and young people with bulimia nervosa — typically 18–20 sessions over 6 months.
- If FT-BN is unacceptable, contraindicated or ineffective, individual eating-disorder-focused cognitive behavioural therapy (CBT-ED) for children and young people with bulimia nervosa may be considered — typically 18 sessions over 6 months, with more frequent sessions early in treatment and up to 4 additional sessions with parents or carers.

46. BDD sertraline 200mg no effect augmentation next step:

- A. Augment sertraline with buspirone,
- B. ECT,
- C. Switch from sertraline to venlafaxine,
- D. Switch from sertraline to another SSRI,
- E. Augment sertraline with mirtazapine.

The table below summarises the NICE Guidelines for the management of OCD and BDD in adults:

Functional impairment	Recommendation
Mild	Low intensity (individual or group) psychological treatments (CBT including ERP). If this is insufficient or can't engage in psychological therapy, then offer choice of either a course of an SSRI or more intensive CBT (including ERP)
Moderate	Either SSRI (any SSRI for OCD but fluoxetine specifically for BDD is first-line) or more intensive CBT (including ERP)
Severe	SSRI + CBT (including ERP)



1.8 Following multidisciplinary review, for adults with BDD, if there has been no response to a full trial of at least 1 SSRI alone, a full trial of combined treatment with CBT (including ERP) and an SSRI, and a full trial of clomipramine alone, the following treatment options should also be considered (note, there is no evidence of the optimal sequence of the options listed below):

- additional CBT or cognitive therapy by a different multidisciplinary team with expertise in BDD
- adding buspirone to an SSRI.

Note that this is an off-label use of buspirone. See [NICE's information on prescribing medicines](#).

47. ICD10 Female presented to A&E sudden aphonia after mother in law coming to live with them-

A. Motor dissociative disorder,

- B. Dissociative seizures,
- C. Acute stress,
- D. Adjustment,
- E. Somatization

[SPMM] A 38-year-old woman presents to A&E with sudden aphonia after her mother-in-law moves in with the woman and her spouse. According to ICD-11, what is the most likely diagnosis?

Select one:

A. Dissociative neurological symptom disorder

- B. Acute stress reaction
- C. Borderline personality disorder
- D. Adjustment disorder
- E. Bodily distress disorder

In ICD-11, dissociative neurological symptom disorder (with speech disturbance) is characterised by symptoms such as difficulty with speaking (**dysphonia**), loss of the ability to speak (**aphonia**), or difficult or unclear articulation of speech (**dysarthria**), attributed to a psychological cause. The initiation or worsening of the symptom or deficit is often preceded by conflicts or other stressors. Symptoms are not produced intentionally.

48. Difference between women & men in bipolar1 men ?

- A. Bipolar men more than women,
- B. Attempt suicide,
- C. Men more common to have first episode mania,
- D. Men higher suicide rates**

[SPMM] Which of the following statements about the diagnostic stability of bipolar disorder is true?
Select one:

- A. The diagnosis of bipolar disorder is often delayed in those with an index depressive phase**
- B. More patients lose than retain a diagnosis of bipolar disorder
- C. Bipolar I to bipolar II switch happens frequently
- D. Bipolar I diagnosis is less stable than bipolar II
- E. After a 5-year symptom-free period, a diagnosis of bipolar can be removed

Explanation: The initial diagnosis of bipolar disorder (most commonly a depressive episode) is often problematic; diagnostic delays of 8-10 years are common. Incorrect diagnoses in patients with bipolar disorder have been reported in 20-60% of cases. Once a diagnosis is made, the stability of bipolar disorder has generally been reported to be high, ranging from 60% to 80%. Bipolar I is more stable than bipolar II simply because bipolar II may switch to bipolar I, but not vice versa. Extreme ages, female sex, psychotic symptoms, changes to treatment, substance abuse, and family history of affective disorder have been related to diagnostic instability.

[SPMM] Lucy and Linda are twin sisters; Lucy has type 1 bipolar while Linda has type 2. Which of the following statements regarding their illnesses is correct?
Select one:

- A. Lucy's diagnosis is more likely to change than Linda's diagnosis
- B. Lucy is more likely to commit suicide than Linda
- C. All of the other listed options are incorrect.**
- D. Lucy will spend more time in depression than Linda
- E. Linda will have better functional recovery than Lucy

Bipolar II disorder is often thought of as a milder expression of bipolar disorder. However, it is only milder in its severity of manic symptoms as hypomania is less severe than mania. Patients with bipolar type II disorder suffer more depressed days than patients with bipolar type I disorder over time. According to prospective studies, bipolar II patients experienced mood symptoms 54% of the time compared with 47% in bipolar I patients. For both bipolar I and bipolar II patients, those mood symptoms were dominated by depressive versus manic symptoms, although bipolar II patients experienced more depressive symptoms (93%) than bipolar I patients (67%). This finding may explain the increased suicidality rates and poor functional recovery observed among bipolar 2 patients

49. Poor prognosis psychosis **Insidious onset**

[SPMM] Miss Thomson is a 24-year-old woman who has been admitted to an inpatient psychiatry unit after beginning to hear voices coming from her light bulbs and having persecutory beliefs about her housemates. Her parents have attended the ward round and are asking what her likely prognosis will be. Which of the following would be the worse prognostic factor in her case?
Select one:

- A. Insidious onset

B. Long Duration of Untreated Psychosis

- C. Immigrant
- D. Female gender
- E. Poor compliance with medication

Features that predict worse outcome for FEP patients include:

- Longer duration of untreated psychosis (DUP).
- Male gender,
- Nonadherence, poor premorbid adjustment.
- The insidious mode of onset,
- The greater severity of negative symptoms,
- Comorbid substance use disorders and
- Having non-affective psychosis.

Prognostic factors in Schizophrenia (with respect to relapses and rehospitalisations)

Good Prognosis	Poor Prognosis
Late onset	Early onset* (not strong)
Obvious precipitating factors	No precipitating factors**
Acute onset	Insidious onset**
Good premorbid adjustment	Poor premorbid adjustment
Affective symptoms (esp. depression)	Social withdrawal
Being married	Single, divorced, or widowed
Family history of affective disorders	Family history of schizophrenia* (weak, controversial)
Good social support	Poor social network, High EE families
Positive symptoms only	Negative symptoms
Good initial response to treatment (best predictor)	Poor compliance
	Neurological signs and symptoms
	History of perinatal trauma
	No remissions in 3 years
	Many relapses
	History of violence

* Suvisaari, JM. Br J Psychiatry. 1998 Dec;173:494-500.

** Rosen, K & Garety, P. Schizophr Bull. 2005; 31:735-750.

Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 9th ed. New York, NY: Lippincott Williams & Wilkins; 2003.

Question 76

Not answered

Marked out of 10.00

Prognosis in schizophrenia

Mr. P has been diagnosed with schizophrenia and his father is keen to know about prognostic options. He is keen to know the answers for the following questions. Please choose the most appropriate responses from the drop-down lists.

- Choose two factors predicting good prognosis in short term
- Choose two factors predicting good prognosis in long term
- Choose two factors predicting poor prognosis in short term
- Choose two factors predicting poor prognosis in long term
- Choose two factors that is not associated with prognostic significance in schizophrenia

Explanation:

In longer term, poor prognosis is predicted by prominent negative (deficit) symptoms, longer duration of initial episode (refer to ISOs data reappraisal by Harrison 2001) and longer duration of untreated psychosis. The issue of family history is unresolved - there is some evidence that family loading of mood disorders may suggest favourable prognosis though this has not been consistently shown. The evidence for poor prognosis in those with positive family history of schizophrenia (non-affective psychosis) is sparse. Age of onset is also controversial when it comes to longer term prediction.

Correct Answer is : 1. Positive response to placebo, Good initial response to antipsychotics 2. Being female, Acute onset of symptoms 3. Daily use of cannabis, Poor compliance with treatment 4. Long duration of untreated psychosis, Male gender 5. Family history of schizophrenia (1st degree relative), Young age of onset

50. Schizophrenia early symptoms in young age:

A. Disorganized behaviour

B. Auditory hallucination

C. Passivity phenomena

D. Paranoid delusion

[SPMM] Compared to adult-onset schizophrenia, early-onset schizophrenia is associated with
Select one:

A. Higher IQ

B. Reduced social deficits

C. Fewer cognitive deficits

D. Increased negative symptoms

E. Better long term outcome

F. Your answer is incorrect.

Explanation: Early-onset schizophrenia is associated with poor premorbid functioning and early developmental delays. Crystallised knowledge and global IQ appear to be more impaired in early-onset schizophrenia cases than in adult-onset cases. **A consistent characteristic is impaired sociability, with significant premorbid difficulties in social development affecting the ability to make and keep friends. Early-onset schizophrenia is characterized by a more insidious onset, negative symptoms, greater disorganisation (incoherence of thought and disordered sense of self), hallucinations in different modalities and, for relatively fewer patients, systematised or persecutory delusions.** It typically has a chronic, unremitting long-term course, with severely impaired functioning in adult life. The long-term outcome is generally poor, but there is heterogeneity.

51. Good response ECT **psychotic depression**

[SPMM] What is the best predictor for response to ECT?

Select one:

A. Psychotic depression

B. Melancholic features

C. Medication responsiveness

D. Severity of depression

E. Personality disorder

Psychotic depression is a clinical predictor of response to ECT. Psychotically depressed patients have been shown to respond better to ECT than severely depressed (without psychosis) and mildly depressed patients. Although the presence of melancholic features appeared to predict a good response to ECT, recent data have challenged this view. The evidence supporting the validity of melancholic features as predictors of response to ECT remains equivocal. Patients without personality disorders have higher responder rates to ECT than patients with personality disorders.

Melancholic feature: Melancholic features of depression typically include:

1. Severe and pervasive sadness or emptiness.
2. Lack of reactivity to pleasurable stimuli (anhedonia).
3. Distinct quality of depressed mood, often worse in the morning.
4. Early morning awakening.
5. Psychomotor agitation or retardation.
6. Significant weight loss or decreased appetite.

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7. Excessive guilt or feelings of worthlessness.
8. Marked cognitive impairment, such as inability to concentrate or make decisions.

52. Affective stupor first line

- A. **lorazepam,**
B. ECT

[SPMM] A 35-year-old woman presents with a 3-day history of mutism and immobility, with no abnormalities found on medical investigations. What is the most appropriate next step of action?
Select one:

- A. **Lorazepam**
B. ECT
C. rTMS
D. Haloperidol
E. Watchful waiting

There are two catatonia subtypes, a **retarded** or **stuporous** form with decreased psychomotor behaviour (as in this case) and an excited form (characterised by agitation, combativeness, impulsivity, and apparently purposeless overactivity). The treatment of the stuporous form is somewhat dependent on its cause but should usually include benzodiazepines. It is postulated that benzodiazepines may act by increasing GABAergic transmission or reducing levels of brain-derived neurotrophic factor. There is most clinical experience with lorazepam. Many patients will respond to standard doses (up to 4mg per day), but repeated and higher doses (between 8mg and 24mg per day) may be needed. When benzodiazepines are effective, the benefit is of rapid onset.

Mr X is a 27-year-old man admitted to a psychiatric unit for treatment of his psychotic illness. His presentation fluctuates. At time, he is restless, agitated and excited. At other times, he is mute, immobile and demonstrates waxy flexibility. Which of the following medications is most likely to help him acutely?

Select one:

- A. **Lorazepam**
B. Risperidone
C. Benzotropine
D. Propanolol
E. Procyclidine

Explanation: This man has catatonia, alternating between excited and stuporous states. In most catatonic patients, inpatient care will be required. **The use of lorazepam (1–2 mg, sublingually, intramuscularly, or intravenously) is recommended,** in addition to treatment of the underlying syndrome. **ECT is also effective in the treatment of catatonia.**

Subtypes of catatonia:

1. **Stuporous/retarded:** associated with signs reflecting a paucity of movement, including immobility, mutism, staring, rigidity, withdrawal, and refusal to eat, along with more bizarre features such as catalepsy, posturing, grimacing, negativism, waxy flexibility, echolalia or echopraxia, stereotypy, verbigeration (repetitive speech, like a scratched record), automatic obedience (exaggerated compliance with commands), and ambitendence (indecisive, hesitant movements)
2. **Excited/delirious:** characterized by severe psychomotor agitation, potentially leading to life-threatening complications such as hyperthermia, altered consciousness, and autonomic dysfunction (“malignant” or “lethal” catatonia)

53. rTMS Absolut CI:

- A. Space occupying lesion ,
- B. Cardiac defibrillator,
- C. Cochlear implant**
- D. Epilepsy,
- E. peacemaker

Contraindications to transcranial magnetic stimulation: rTMS should not be administered to patients who have **ferromagnetic or magnetic sensitive metal objects** implanted in the head or neck areas in close proximity to the TMS coil magnetic fields

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5846193/#:~:text=Contraindications%20to%20transcranial%20magnetic%20stimulation,the%20TMS%20coil%20magnetic%20fields>

Repetitive transcranial magnetic stimulation Oxford handbook

Adverse effects: Minimal, but patients often report headache or facial discomfort; rarely, seizure induction. Indications Experimental treatment for treatment- resistant depression;

possible use in treatment of treatment- resistant auditory hallucinations; negative symptoms of schizophrenia; OCD; panic disorder.

Contraindications: **History of stroke, brain tumor, or epilepsy.**

54. Difference between BPD, Autism **fear of abandonment**

[SPMM] Which of the following is a characteristic feature of borderline personality disorder?
Select one:

- A. Attention seeking behaviour
- B. Fear of abandonment**
- C. Avoidance of social contact
- D. Generalised lack of trust
- E. Fear of boredom

Explanation: Features of borderline personality disorder (5/9 required for DSM-5 diagnosis):

- ➔ Unstable self-image
- ➔ Unstable personal relationships
- ➔ Impulsive behaviour in areas such as personal safety and substance misuse
- ➔ Self-harm, suicidal threats or gestures
- ➔ Affective instability due to marked reactivity of mood
- ➔ Episodes of micro-psychosis
- ➔ Chronic feelings of emptiness

55. Most common comorbidity on gender dysphoria

- A. Mood disorder**
- B. Personality

Major depressive disorder (33.7%), specific phobia (20.5%), and adjustment disorder (15.7%) were the three most prevalent disorders. Conclusion. Consistent with most earlier researches, the majority of patients with gender dysphoria had psychiatric Axis I comorbidity.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4142737/>

Transgender populations have poorer health outcomes than the general population, with an increase prevalence of **affective, anxiety, and substance misuse disorders**. There are conflicting data regarding any difference in the prevalence of personality disorders. A Swedish long- term cohort study has shown that high rates of premature mortality, suicidal behaviour, and psychiatric comorbidity exist, even after treatment for gender dysphoria. It remains impractical to design long- term prospective, controlled studies comparing psychiatric outcomes in transgender individuals who do and do not receive treatment. Transgender women are regarded as being a high- risk group for HIV and other sexually transmitted infections (STIs). Other sexual and reproductive health issues affecting transgender women and men are not well studied.

56. Cyclothymia duration **2 years**

[ICD11] Cyclothymic disorder is characterized by a persistent instability of mood over a period of at least **2 years**, involving numerous periods of hypomanic (e.g., euphoria, irritability, or expansiveness, psychomotor activation) and depressive (e.g., feeling down, diminished interest in activities, fatigue) symptoms that are present during more of the time than not.

Subtypes of bipolar disorder
Select one diagnostic option for each of the following scenarios

A patient, who has been diagnosed with depression and was started on venlafaxine, now presents a history of feeling high with 'unstoppable energy'.	Choose...
A patient who is known to have mood disturbances is taking lithium. In the last 6 months he has had one episode each of low mood and mania, followed by another episode of high energy, sleep loss and euphoric mood. After an intervening period of normal mood, he developed another episode of mania.	Choose...
A 33-year-old man presents with a history of numerous periods with hypomanic symptoms and numerous periods with depressive symptoms in last 2 years that do not meet criteria for a mania or a major depressive episode.	Choose...
A 25-year-old woman has episodes of low mood and irritability. She described her energy levels to be high and is sleeping only for 4 hours a day.	Choose...

Your answer is incorrect.

Explanation:

Case 1 can be described as an example of bipolar 3 disorder; it is a variant of bipolar disorder with predominant depression complicated by antidepressant induced hypomania. This presentation is a feature of the bipolar spectrum (in ICD: this is coded as unspecified type).

Case 2 is suggestive of rapid cycling, where at least four episodes of bipolar disorder occur within a period of one year. 70-80% of rapid cyclers are women. Some of the factors associated with the rapid cycling include the use of tricyclic anti-depressant, low thyroxine level, being a female patient and having bipolar 2 rather than type 1 illness. Ultra-rapid cycling refers to the situation when fluctuations are over days or

Case 3 is suggestive of cyclothymia in which an individual may experience oscillating high and low moods (numerous periods of mild depression and mild elation), without ever having a manic or depressive episode that is sufficiently severe or prolonged to fulfil the criteria for BPAD or recurrent depressive disorder. The mood swings are usually as being unrelated to life events.

Mixed states are instances where manic and depressive symptoms occur simultaneously, as seen in case 4 above. The occurrence of both manic/hypo manic and depressive symptoms in a single episode, present every day for at least 1 week (DSM-5) or 2 weeks (ICD-10)

The correct answer is: A patient, who has been diagnosed with depression and was started on venlafaxine, now presents a history of feeling high with 'unstoppable energy'. → Bipolar disorder, unspecified, A patient who is known to have mood disturbances is taking lithium. In the last 6 months he has had one episode each of low mood and mania, followed by another episode of high energy, sleep loss and euphoric mood. After an intervening period of normal mood, he developed another episode of mania. → Rapid cycling bipolar disorder, A 33-year-old man presents with a history of numerous periods with hypomanic symptoms and numerous periods with depressive symptoms in last 2 years that do not meet criteria for a mania or a major depressive episode. → Cyclothymia, A 25-year-old woman has episodes of low mood and irritability. She described her energy levels to be high and is sleeping only for 4 hours a day. → Mixed affective state

57. At mental state prevent 1st episode psychosis, What next step ?

- A. **Antipsychotic,**
- B. Statin prevent progression

We are not sure what the questions is asking but if about type of prevention → Primary indicated.

[SPMM] A 34 year old woman has acute onset of intense suspiciousness and passivity phenomenon with auditory hallucinations. With respect to the evidence based treatment of first episode psychosis, which of the following is true?

Select one:

- A. There is evidence in favour of lower antipsychotic doses**
- B. First generation antipsychotics are no better than placebo
- C. Choice of an antipsychotic should be based on the gender of the patient
- D. Second generation antipsychotics are no better than placebo
- E. RCTs show a clearly significant difference between clozapine and other antipsychotics

Explanation: Low to moderate dosage of antipsychotic monotherapy is the first line of treatment for first-episode psychosis. Doses should be well below those for established illness. Approximate, typical, daily mean doses for common antipsychotic medications are as follows: risperidone 2- 4mg, haloperidol 2-5mg, quetiapine 400-500mg, amisulpride 450mg, and olanzapine 10mg. Higher doses generally increase the risk of side effects with limited additional therapeutic benefit in first-episode patients. There is no clear indication from well-designed, double-blind RCTs or well-conducted meta-analyses that any antipsychotic medication is more efficacious than others for first-episode psychosis. Both first and second generation antipsychotics have been shown to be efficacious.

A small number of studies have examined clozapine as a first- or second-line treatment modality in schizophrenia spectrum disorders. Current evidence shows that as a first- or second-line treatment, clozapine outperforms other antipsychotics. When including only RCTs, beneficial effects of clozapine over other antipsychotics become insignificant, although the direction of effect (clozapine outperforming other antipsychotics) remains unchanged.

There is evidence to suggest that women have a better response to medication early in their course of treatment and require lower doses of antipsychotics, though this benefit appears to dissipate with advancing age. Hormonal factors, including changes with child-bearing and menopause, have been proposed but this remains an area of investigation.

58. Best universal prevention against alcohol dependence?

- A. Increase price,**
- B. Education in schools

59. Nolan Q **how change improvement**

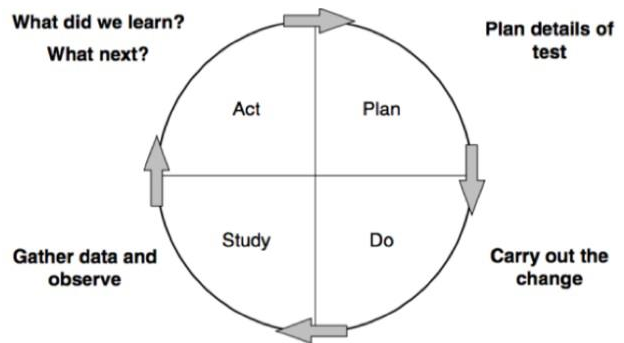
[SPMM] Thomas Nolan described a quality improvement method, which involved three central questions. Which of the following is one of these central questions?

Select one:

- A. What are we trying to accomplish?**
- B. How much do you want to save?
- C. Who will it affect?
- D. What measurements will we use?
- E. What do we want to know?

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The PDSA cycle was initially developed by Tom Nolan and colleagues. The PDSA Model for Improvement consists of: three fundamental questions; the plan-do-study-act (PDSA) cycle; and six areas of knowledge. To make improvements, first ask: What are we trying to accomplish? This will focus the efforts. Then ask: How will we know that a change is an improvement? Identifying criteria and collecting data will help. The final question is: What changes can we make that will result in improvements? This requires testing, especially using the PDSA cycle. See the attached picture)



Models and Tools for Quality Improvement

For each of the questions regarding quality improvement below, choose one or more appropriate answers as indicated from the list above.

1. An improvement system used to develop new processor products at superior performance levels. (Choose ONE)

✗

2. An extension of PDSA model describing events and questions that must precede the initiation of a PDSA cycle (Choose TWO)

✗ ✗

3. A visualisation tool to capture variations that occur within a system (Choose ONE)

✗

Benefit Realisation Plan
DMAIC model
DMADV model
FADE model
Failure Mode and effects analysis
FOCUS approach
Lean Thinking
Model for Improvement (MFI)
Pareto Chart
PDCA cycle
PDSA cycle
Six Sigma Model
Statistical Process Chart

Benefit Realisation Plan is a table used as a tool to ensure that the intended benefits originally planned in a QI project are actually delivered to the stakeholders in a timely fashion. The FOCUS approach to QI is shown in the figure below. FOCUS stands for Finding a process to improve, Organising a team, Clarifying current knowledge, Understanding causes of variation and Selecting the process improvement procedures. Model for improvement (MFI) is commonly employed now by many NHS organizations. MFI is an extension of PDSA and describes reflections that need to occur before embarking on a PDSA cycle. To measure outcomes over time, tools are available to generate time series graphs or frequency charts. Commonly used time series graphs include 'run charts' or Statistical Process Control charts (SPC). SPC enables the use of simple statistical concepts to understand processes that are being studied or implemented. In essence, SPC helps to assess the 'system' for predictability of outcome when a change is implemented; if unexpected variations are present, this provides an opportunity to study the external factors that may influence the desired outcome.

DMADV (Define-Measure-Analyze-Design-Validate) is a Six Sigma framework that focuses primarily on the development of a new service, product, or process as opposed to improving a previously existing one. It is especially useful when implementing new strategies and initiatives because of its basis in data, early identification of success, and thorough analysis. (Ref: <https://www.sixsigmadaily.com/what-is->

Answers: 1. DMADV model, 2. FOCUS approach, Model for Improvement (MFI), 3. Statistical Process Chart

60. How many weeks needed to decrease and stop mood stabilizer after bipolar patient become stable

A. 4 weeks,

B. 2 weeks,

C. 12 weeks

[SPMM] When used as mood stabilizers, what is the minimum period suggested to withdraw anticonvulsant drugs?

Select one:

A. 3 months

B. 1 month

C. 1 week

D. 1 year

E. 6 months

Explanation: It is not known if the abrupt discontinuation of anticonvulsants (such as valproate and carbamazepine) worsens the natural course of bipolar illness in the same way that discontinuation of lithium does. Until further data are available, it is recommended that anticonvulsants should be discontinued slowly, over at least a month.

61. Post natal depression in developed countries

A. 5-10%,

B. 11-15%

[SPMM] Post-natal depression can lead to significant distress in new mothers and impair their ability to bond with and care for their newborn children. What percentage of new mothers suffers from postnatal depression?

Select one:

A. 10-15%

B. 0-5%

C. 15-20%

D. 5-10%

E. 20-25%

It is thought that 10-15% of new mothers experience post-natal depression, with rates reducing to <10% after 12 weeks postpartum

Old

62. Alzheimer risk of when your sibling, parent has AD to general population

A. 20%,

B. 10%,

C. 30%,

D. 40%

[SPMM] Lucy's father has been diagnosed with Alzheimer's disease in his 80s. What is the increased risk for her to develop Alzheimer's disease when compared to someone without a positive family history?

Select one:

A. 3 times risk

B. 10 times risk

C. 5 times risk

D. 20 times risk

E. 2 times risk

An actual predicted risk of developing Alzheimer's disease in the first-degree relatives of probands with Alzheimer's disease is 15-19%, compared with 5% in controls. "Thus, the risk to the first-degree relatives of patients with Alzheimer's disease who developed the disorder at any time up to the age of 85 years is increased some **3 – 4 times** relative to the risk in controls. This would seem to translate to a risk of developing Alzheimer's disease of between one in five and one in six".

- ➡ **People homozygous for the E4 allele have between a 10 and 30 times increased risk for Alzheimer's compared to people not carrying any E4 alleles.**
- ➡ **Heterozygotes for APOE4 have a 3 times increased risk. In addition each E4 allele lowers the age of onset by 10 years.**
- ➡ **APOE4 found in >50% of people with AD**

63. AD different from normal aging in stairs
Concentration

The risk of developing Alzheimer's disease significantly increases with age, especially after the age of 65. While the exact multiplier for risk increase from age 65 to 85 can vary depending on the study and population, the consensus in the scientific community is that the risk approximately doubles every 5 years after age 65. Thus, from age 65 to 85 — a span of 20 years — the risk of developing Alzheimer's disease increases substantially.

If we consider the doubling rule:

- From 65 to 70, the risk doubles once.
- From 70 to 75, it doubles again (2 times 2 = 4 times the original risk by age 70).
- From 75 to 80, another doubling (4 times 2 = 8 times the original risk by age 75).
- From 80 to 85, yet another doubling (8 times 2 = 16 times the original risk by age 80).
- Therefore, by the time an individual progresses from age 65 to 85, the risk of developing Alzheimer's disease has increased by approximately 16 times compared to their risk at age 65.

64. Granulin gene in any type of dementia **FTD**

[SPMM] Frontotemporal dementia (FTD) is a highly heritable group of neurodegenerative disorders, with around 30% of patients having a strong family history. Most of that heritability is accounted for by autosomal dominant mutations in the chromosome 9 open reading frame 72 (**C9orf72**), progranulin (**GRN**), and microtubule-associated protein tau (**MAPT**) genes, with mutations more rarely seen in several other genes. Behavioural variant FTD (bvFTD) is the most common diagnosis in each of the genetic groups, although in C9orf72 carriers, amyotrophic lateral sclerosis either alone, or with bvFTD, is also common.

65. MOCA scale 22 then 11 after week apart

- A. **delirium**,
- B. dementia

66. Parkinson, psychosis best evidence treatment:

- A. **Clozapine.**
- B. Quetiapine
- C. Rivastigmine

Treatment of dementia
What would be the most appropriate treatment for?

A 74-year-old man with a diagnosis of Alzheimer's disease who presents with severe behavioural problems and is not responding to donepezil.

A 78-year-old woman with a diagnosis of Alzheimer's disease who is taking donepezil and presents with a month's history of low mood, poor appetite, weight loss and insomnia. Her sister recently died from cancer.

An 80-year-old man with a diagnosis of Parkinson's Disease who develops significant cognitive impairment

A 92-year-old woman who has had a step-wise deterioration of cognition, which appears to coincide with historic strokes.

A 73-year-old male on Sinemet therapy who experiences improvement in his Parkinsonian symptoms but within weeks complains of seeing goblins in the evening

Choose...

- Memantine
- Mirtazapine
- Donepezil
- Risperidone
- Aggressive treatment of risk factors
- Rivastigmine
- Quetiapine
- Ginkgo Biloba
- Aspirin

Explanation:

- 1) **Memantine**, an NMDA-antagonist would be an appropriate choice for patients who are not responding to donepezil, an acetylcholinesterase inhibitor, and who present with challenging behaviour. It is important to check the renal function (including eGFR) of any patient prior to commencing memantine as it is eliminated by the kidneys and can cause renal impairment
- 2) This lady is presenting with symptoms of depression and would benefit from treatment with antidepressant medication. Mirtazapine is a popular choice for antidepressants in the elderly in view of its sedative effects (for those suffering from insomnia) and good side effect profile.
- 3) **Rivastigmine**, an acetylcholinesterase inhibitor, has been evaluated in the symptomatic treatment of patients with mild-to-moderate dementia associated with idiopathic Parkinson's disease. Although the evidence is preliminary, it is the best available evidence among the other options for Parkinson's related dementia. Rivastigmine appears to improve both cognition and activities of daily living in patients with PDD, resulting in a clinically meaningful benefit in a large number of cases.
- 4) This lady is suffering from vascular dementia. Anti-dementia drugs (acetylcholinesterase inhibitors) are not currently licensed for use in vascular dementia, although some studies suggest a possible benefit.
- 5) In patients with Parkinson's Disease who develop psychotic symptoms following treatment with dopamine agonists, use oral atypical antipsychotic medication, preferably quetiapine, (12.5mg-75 mg but higher doses may be required) as they are associated with lower incidence of EPSEs. Furthermore, they do not require blood count surveillance. **In refractory cases use clozapine in small doses as it is the most effective and only licensed antipsychotic in PD.** Regular blood tests and close monitoring for neutropenia and agranulocytosis is essential in clozapine therapy.

67. Parkinson auditory hallucination, cognitive impairment MMSE 22/30 **Rivastigmine**

If only psychosis without cognitive impairment → antipsychotic

[SPMM] Mrs Lisbeth Slander is a 79-year-old woman who has progressively developed loss of memory, visual hallucinations and intermittent irritability over the course of several months. Which of the following would be the most appropriate treatment for her condition?
Select one:

- A. Rivastigmine**
- B. Memantine
- C. Diazepam
- D. Fluoxetine
- E. Amitriptyline

This is most likely to represent Lewy Body Dementia, which is characterized by memory impairment, perceptual disturbances and mood changes. The first line of treatment is acetyl cholinesterase inhibitors, such as Rivastigmine.

68. DLB probable core feature:

- A. REM behavioral disorder ,**
- B. REM behavior disorder without atonia,
- C. Hallucination of any Modality,
- D. SPECT decrease dopamine in basal ganglia,
- E. Sensitivity neuroleptic

Core clinical features of dementia with Lewy bodies (DLB): REM-PCV

1. Fluctuating Cognition with pronounced variations in attention and alertness
2. Recurrent Visual hallucinations that are typically well formed and detailed
3. REM sleep behaviour disorder, which may precede cognitive decline
4. Spontaneous cardinal features of Parkinsonism (bradykinesia, rest tremor, and/or rigidity)

Supportive clinical features of DLB:

1. Severe sensitivity to antipsychotic agents
2. Postural instability Repeated falls
3. Syncope or other transient episodes of unresponsiveness
4. Severe autonomic dysfunction (e.g. constipation, orthostatic hypotension, urinary incontinence)
5. Hypersomnia
6. Hyposmia
7. Systematized delusions
8. Hallucinations in other modalities
9. Apathy, anxiety, and depression

Forensic

69. Automatism

A. Loss of consciousness 1-2 minutes,

B. Must precede aura

[SPMM] Which of the following is classified under sane automatism?

Select one:

- A. Epilepsy
- B. Sleepwalking
- C. Sleep terrors**
- D. Brain tumours
- E. All of the listed options

Explanation: If an individual commits an offence when their body is not under the control of their mind, (e.g. when asleep), they are not guilty of the offence. Legally, this is called an automatism. In England and Wales, two legal types of automatism are recognised: insane and sane (automatism simpliciter). The distinction is based on whether the behaviour is likely to recur. Insane automatisms are due to an intrinsic cause (e.g. sleepwalking, brain tumours, epilepsy) and result in acquittal on the grounds of insanity. Sane automatisms are due to an extrinsic cause (e.g., **confusional states, concussion, reflex actions after bee stings, dissociative states, sleep terrors, and hypoglycaemia**) and result in complete acquittal.

70. Forensic psychiatry conviction, remand, sentence, trial.

Remand typically refers to the process of detaining an accused person in custody until their trial or until further legal action is taken. It can also refer to sending a case back to a lower court for further consideration or a new trial. In the context of criminal law, remand often occurs when a defendant is deemed a flight risk or a danger to the community, or when there is concern that they may tamper with evidence or obstruct justice if released before trial.

A trial is a formal legal proceeding in which the facts of a case are examined, witnesses are heard, and a decision is made by a judge or jury. It is typically conducted to determine the guilt or innocence of a defendant accused of committing a crime. During a trial, both the prosecution, representing the state or the plaintiff, and the defense present evidence, question witnesses, and make arguments to support their

case. The judge or jury then deliberates and renders a verdict based on the evidence presented and the applicable law.

A **conviction** in legal terms refers to the formal declaration by a court that someone is guilty of a criminal offense. It occurs after a trial or when the accused person pleads guilty. A conviction typically results in a punishment, such as a fine, imprisonment, probation, or other penalties, depending on the severity of the offense and the laws of the jurisdiction.

In legal contexts, a **sentence** is the punishment imposed by a court on someone who has been convicted of a crime. It can include various forms of punishment, such as fines, community service, probation, or imprisonment. The length and type of sentence depend on factors such as the severity of the crime, the defendant's criminal history, and any mitigating or aggravating circumstances presented during the trial.

Arrest > Police stn > court > remand > conviction > prison

71. Prisoner vs prisoner released from hospital:

- A. less likely to reoffend,
- B. homeless,
- C. more likely to commit suicide,
- D. higher mortality

72. Suicide in prison:

A. Suicide in prisoner more than general population,

- B. Overcrowded prisons,
- C. Lower education of staff.

[SPMM] Internationally, which of the following is true of suicide in prison?

Select one:

A. Repetitive self-harm increases suicide risk

- B. Suicide is less common in prison than in the general population
- C. Low staff engagement and education explain increased suicide risk
- D. Overcrowding consistently increases the risk of suicide
- E. The majority of suicides are in women

Prison suicide is an international problem, and **rates of suicides in prisoners are higher than in general populations. Psychiatric illness, substance misuse, and repetitive self-harm are known important risk factors for suicide.** The majority of suicides are in men (>90%), but women are at particular risk (in female prisoners, rate ratios are 20 times higher than in the general population). Findings of an association between prison suicide and overcrowding are inconsistent. A number of studies have found a positive correlation with overcrowding and prison suicide, but other have found the opposite, possibly due to a protective effect of the double occupation of cells designed for single use. Low staff engagement and education, negative staff-prisoner relationships, and high prison population turnover rates are also potential explanations, but evidence is limited.

[SPMM] Of those listed, the greatest risk factor for suicide in a prisoner is:

Select one:

- A. Female sex
- B. Elderly age
- C. History of substance misuse**
- D. History of arson

E. History of sexual offences

Explanation: Prison suicide and self-harm has been rising in the UK. The most common method of suicide in prison is hanging. The majority of suicides occur within 6 months of imprisonment; life-sentence prisoners are over-represented. Occupying a single cell/solitary confinement is a particularly strong risk factor for both self-harm and suicide. Remand prisoners, young offenders, and those with a history of substance misuse and violent offences are also at higher risk of suicide. And being married

73. Disorder most common in those who commit arson **alcohol use**

[SPMM] With respect to arson, which of the following is correct?

Select one:

- A. Males offend more than females
- B. Alcohol misuse is very common
- C. Women receive disproportionately more psychiatric disposals
- D. There is an association with low intellectual functioning

E. All of the listed options

Explanation: Arson is a criminal act in which one willfully and maliciously sets fire to or aids in setting fire to a structure, dwelling, or property of another. Most arsonists are male, aged late adolescence-early adulthood, unmarried, poorly educated, living alone, and unemployed or unskilled labourers. Psychiatric disorders are over-represented in arsonists compared to the general population and other offender groups. Substance use disorders (especially alcohol) and personality disorders (particularly antisocial) are most common. Others include psychotic disorders and intellectual disability/low intellectual functioning.

74. Victims of female homicide if has mental disorder belong to the age group **<16**

[SPMM] Victims of female homicide perpetrators with psychosis often belong to the age group of;

Select one:

- A. 60-75
- B. <16**
- C. 16-35
- D. >75
- E. 36-60

The victims of female homicide perpetrators are often in their own family. A substantial number of these victims are <16 years of age, especially if the perpetrator has psychosis. 50% of mentally ill perpetrators of child homicide have a psychotic disorder. But note that fathers are significantly more likely to kill their children than mothers, if one considers all child homicides. Also 13% of perpetrators of filicide (child killing by parents) took their own life after killing their child (this is called filicide-suicide)

[SPMM] In females, which of the following age group is associated with the highest risk of committing a homicide?

Select one:

- A. 40-49
- B. <16
- C. 20-29
- D. 30-39**
- E. 16-19

Over 40 per cent of female perpetrators of intimate partner homicide are aged between 30 and 39' - the average age being 35 years.

75. Unfit to plead amnesia for event hallucination delusion for crime thought

A person's fitness to plead is judged by specific criteria:

Pritchard criteria (case law 1836)
Must be able to instruct his solicitor and counsel
Must understand the charge(s)
Must be able to decide whether to plead guilty or not
Must be able to follow court proceedings enough to make a proper defence
Must know they can challenge a juror
Must be able to give evidence in his or her own defence

[SPMM] Which of the following defendants is most likely to be fit to plead in a court trial?
Select one:

- A. A defendant who cannot understand the nature of the charges
- B. A defendant with amnesia for the offence**
- C. A defendant who does not understand trial procedures
- D. A defendant with lack of ability to challenge the jurors
- E. A defendant who is catatonic, mute and not able to communicate

Explanation: R v Pritchard criteria are often cited when discussing the 'test' for fitness to plead. An individual is unfit to plead if they are unable to:

- ➔ Understand the course of proceedings on trial
- ➔ To comprehend the details of evidence
- ➔ To know they might challenge any of the jurors to whom they might object
- ➔ To instruct legal counsel

[Psych-Mentor] Which of the following conditions accounts for the most cases of 'unfit to plead'?

- A. Antisocial personality disorder
- B. Learning disability
- C. Depression
- D. Dementia
- E. Schizophrenia**

Stats

76. Type 1 error

- A. Rejecting null hypothesis when it's true**
- B. Null hypothesis accepted when it's false, rejected when it's false
- C. Null hypothesis accepted when it's true

Error Type	Description	Symbol
Type 1 Error <i>False Positive</i>	Incorrectly rejecting a null hypothesis when it's true.	α (Alpha)
Type 2 Error <i>False Negative</i>	Incorrectly failing to reject a null hypothesis when it's false.	β (Beta)

[SPMM] Assigning the mean of completed observations in place of missing data while carrying out intention to treat analysis is called mean imputation. This increases the likelihood of
Select one:

- A. Type 2 error
- B. Obtaining imprecise results
- C. Type 1 error**
- D. Confounding bias
- E. Measurement bias

By using a standard mean value in place of widely variable missing values, a false precision is injected into the observations. This might produce false positive result i.e. type 1 error.

➤ **Type 2 error is more dangerous than type 1 error**

Statistical Concepts and Definitions

You are reviewing research papers in the field of psychiatry. Each paper mentions a statistical concept or term. Match the statistical concepts or terms below with correct descriptions. Each term can be used more than once or not at all.

1. The probability of making an incorrect rejection of a null hypothesis when it is true.

✗

2. The probability of correctly detecting a true effect or difference in a study.

3. The level of statistical significance used to determine whether to reject the null hypothesis.

✗

4. The probability of failing to detect a true effect or difference when it exists.

✗

5. A measure of the likelihood that the results of a study occurred by chance.

✗

p-value
Power
Type 1 Error
Type 2 Error
Significance Level (alpha)
Type 3 error
Heterogeneity

CORRECT ANSWERS:

1. The probability of making an incorrect rejection of a null hypothesis when it is true. - Type 1 Error
2. The probability of correctly detecting a true effect or difference in a study. - Power
3. The level of statistical significance used to determine whether to reject the null hypothesis. - Significance Level (α)
4. The probability of failing to detect a true effect or difference when it exists. - Type 2 Error
5. A measure of the likelihood that the results of a study occurred by chance. - p-value

EXPLANATION:

- The probability of making an incorrect rejection of a null hypothesis when it is actually true is referred to as "Type 1 Error". This error occurs when a statistical test incorrectly indicates a significant result when there is no real effect or difference.
- The probability of correctly detecting a true effect or difference in a study is known as "Power". Power represents the ability of a statistical test to correctly identify a significant result when a true effect exists. Higher power indicates a better chance of detecting real effects.
- The level of statistical significance used to determine whether to reject the null hypothesis is referred to as the "Significance Level (α)". Commonly used significance levels include 0.05 (5%) and 0.01 (1%).
- The probability of failing to detect a true effect or difference when it exists is known as "Type 2 Error". This error occurs when a statistical test fails to identify a significant result when there is a real effect. Type 2 Error is the complement of Power (i.e., $1 - \text{Power} = \text{type 2 error rate}$).
- A measure of the likelihood that the results of a study occurred by chance is represented by the "p-value". The p-value quantifies the probability of obtaining observed results (or more extreme results) if the null hypothesis is true. Smaller p-values suggest stronger evidence against the null hypothesis.

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77. cofounder decrease **Stratification analysis**

Sampling methods

Choose one option for each question below.

A researcher is interested in measuring dimensional factors of PTSD in illegal immigrants. He could not find a reliable sampling frame for recruitment.

Choose...

A researcher is interested in studying the lifestyle factors influencing metabolic syndrome in patients who receive antipsychotic prescriptions. He classifies the eligible sample into inpatients, outpatients and CMHT clients before recruiting for his study, to balance for the effect of one's environment on metabolic syndrome.

Choose...

A research team is interested in studying the prevalence of depression across all primary care attendees in the UK. But due to the scope of the task, they decide to pick randomly a set of representative primary care authorities and survey all patients in the selected regions.

Choose...

A research nurse wants to study the experience of some mentally ill parents with respect to the demands of their teenage children. She designs a qualitative study method to carry out this work.

Choose...

Your answer is incorrect.

Explanation:

Study 1: In snowball sampling, one participant identifies many further potential subjects, creating an avalanche (snowball) effect.

Study 2: **Stratification** has been done in this study.

Study 3: In cluster sampling, the units of examination are whole groups rather than individuals.

Study 4: More accurately this must be purposive sampling. This is a form of convenience sampling.

[SPMM] Which of the following methods can be used to evaluate both confounders and effect modifiers in a study examining causal association

Select one:

- A. Standardisation
- B. Minimisation
- C. Randomisation
- D. Restriction
- E. Stratification**

Stratification can help analyses the effect modifiers and confounders in a study.

[SPMM] Both known and unknown confounders can be controlled by which of the following methods?

Select one:

- A. Matching
- B. Blinding
- C. Randomisation**
- D. Stratification
- E. Restriction

Randomisation can control both known and unknown confounder variables. Blinding is not a method for controlling confounders.

78. Not normally distributed **Mann whitney**

Non-parametric tests

Identify the names of statistical tests described below

This test is the non-parametric equivalent of the paired t-test

This test is the non-parametric equivalent of the 2 sample t-test

This test is the non-parametric equivalent of the one-way ANOVA

A non-parametric test used for a 2 x 2 or larger table.

Choose...

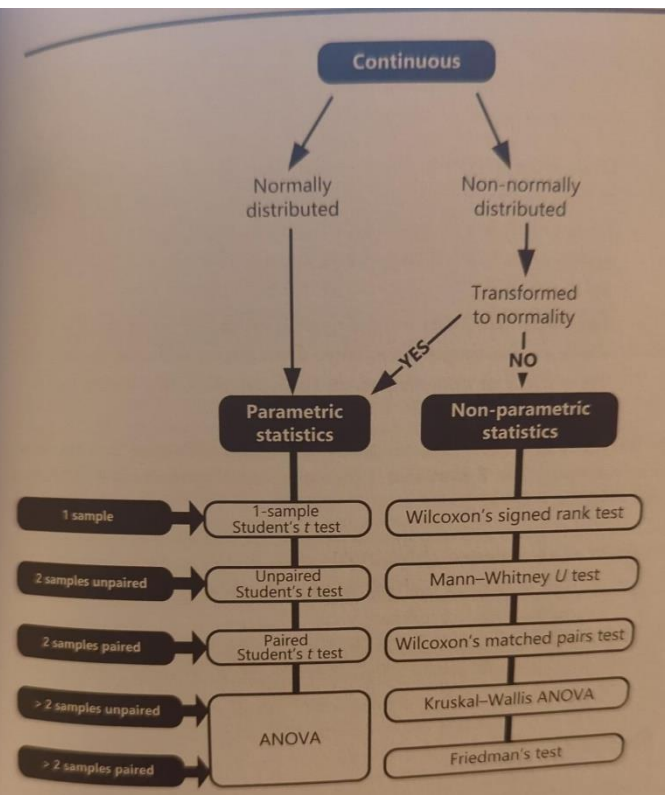
1 sample (paired) t-test
Mann-Whitney U test
Chi-squared test
Wilcoxon signed rank test
Two-way ANOVA
Area under a curve
One way ANOVA (analysis of variance)
ANCOVA (analysis of covariance)
Kruskal-Wallis test
2 sample (unpaired) t-test

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- 1) Wilcoxon signed rank test (*paired/married man coaxed with his partner*) is the non-parametric equivalent of the t-test comparing paired observations.
- 2) Mann-Whitney U test (**U=Unpaired**) and is the non-parametric equivalent of the unpaired, 2-sample t-test, comparing the distributions two independent groups of observation.
- 3) Kruskal-Wallis (*man going through crucial time he is unpaired*) test is the non-parametric equivalent of the one-way ANOVA, which is used to compare the distributions of more than two independent groups of observation.
- 4) The chi-square test is a non-parametric test used for a 2 x 2 or larger table. Fisher's exact test is similar to chi-square test but used for smaller sample sizes

- ➔ **Friedman** = paired man fried daily by his partner
- ➔ **Chi-square** = any 2X2 table

Statistical Test	Use	Assumptions	Example Use Case
t-Test	Compare means of two groups	Normally distributed data, equal variances	Compare test scores of two teaching methods
ANOVA	Compare means of more than two groups	Normally distributed data, equal variances	Compare test scores among three teaching methods
Chi-Square Test	Test independence between categorical variables	Random sample, Large enough sample size	Test if there's an association between gender and preference for tea or coffee
Pearson Correlation	Measure linear association between two continuous variables	Linearity, Homoscedasticity	Examine the relationship between age and income
Regression Analysis	Predict a continuous dependent variable based on one or more independent variables	Linearity, Homoscedasticity, Independence of errors	Predicting house prices based on square footage, number of bedrooms, etc.
Mann-Whitney U Test	Compare distributions of more than two independent groups	Independence of observations	Compare test scores between two different schools
Kruskal-Wallis Test	Compare distributions of more than two independent groups	Independence of observations, Similar distributions	Compare performance of different teaching methods across multiple schools
Wilcoxon Signed-Rank Test	Compare distributions of two related groups	Dependent variable should be continuous or ordinal	Compare pre-test and post-test scores within a group
McNemar's Test	Compare paired proportions or frequencies in a 2x2 contingency table	Binary data, Dependent groups	Compare the effectiveness of two treatments on a binary outcome
Fisher's Exact Test	Compare proportions in a 2x2 contingency table	Low sample size (< 20), Independence of observations	Assess the association between gender and smoking status in a small sample



	Comparison				Association	Regression
	2 data sets		>2 data sets		Correlation between 2 variables	Predicting one from another
	Paired	Un-paired	Paired	Un-paired		
Normal Distribution (Mean)	Paired t test	Un-paired t test	Repeated measure ANOVA	One way ANOVA	Pearson correlation	Linear regression
Non - Normal Distribution (Median)	Wilcoxon signed-rank test	Wilcoxon rank sum test Mann-Whitney U test	Friedman test	Kruskal-Wallis test	Spears Rank correlation	Non Parametric Regression
Dichotomous data	McNemar's Test	Chi-square test Fisher's exact test	Cochran's Q test	Chi-square test	Contingency Coefficient	Logistic Regression

79. Risk ratio equal Hazard ratio

A. When prevalence low,

B. Hazard ratio depend on time

Cox's proportional hazards regression

The **hazard rate** is the probability of an endpoint event in a time interval divided by the duration of the time interval. The hazard rate can be interpreted as the instantaneous probability of an endpoint event in a study if the time interval is short. The hazard rate might not be constant during the study.

The **hazard ratio** is the hazard rate in the experimental arm divided by the hazard rate in the control arm. The hazard ratio is complemented by a *P* value and confidence intervals. It is assumed that the hazard ratio remains constant throughout the study. **SEE EXCERPT 69**

If the hazard ratio is 1, the two arms have an equal hazard rate.

If the hazard ratio is >1 , the experimental arm has an increased hazard.

If the hazard ratio is <1 , the experimental arm has a reduced hazard.

Although the interpretation of hazard ratio is similar to that of relative risk and the odds ratio, the terms are not synonymous. The hazard ratio compares the experimental and control groups throughout the study duration, unlike the relative risk and odds ratio, which compare only the proportion of subjects that achieved the outcome in each group at the end of the study.

It is not possible to calculate the hazard ratio from a 2×2 contingency table.

Cox's proportional hazards regression is used to produce the hazard ratio. It is the multivariate extension of the log-rank test. It is used to assess the impact of treatment on survival or other time-related events and adjusts for the effects of other variables.

[SPMM] Which of the following assumptions must be met before applying Cox's proportional hazard test? Select one:

- A. The survival curves cross each other during the period of follow-up
- B. The median survival time is reached during the period of follow-up
- C. The hazard ratio is constant over time
- D. At least 50% of the observed events must have occurred in the first year of observation
- E. Confounders have no effect on survival time

We can use either log-rank test or Cox test to determine statistically the difference between two survival curves. The log-rank test is used to test whether there is a difference between the survival times of different groups, but it does not allow other explanatory variables to be taken into account. Cox's proportional hazards model is analogous to a multiple regression models and enables the difference between survival times of particular groups of patients to be tested while allowing for other factors. In this model, the response (dependent) variable is the 'hazard'. The hazard is the probability of dying (or experiencing the event in question) given that patients have survived up to a given point in time, or the risk for death at that moment. In Cox's model no assumption is made about the probability distribution of the hazard. However, it is assumed that the ratio of the risk for dying at a particular point in time between one group and the other is constant all the time. In other words, Cox test assumes that the hazard ratio does not depend on time.

[SPMM] An investigator is interested not only in the total number of new-onset hyperlipidaemia following antipsychotic use, but also in the timing of onset after initiating the treatment. So he attempts to calculate an average measure of the relative risk weighted for number of patients at risk over a long period of follow-up in the trial cohort. This measure is best described as
Select one:

- A. Effect size
- B. Odds ratio
- C. Hazard ratio**
- D. Standardised mean difference
- E. Degree of freedom

The weighted RR over the entire study in a survival analysis is known as the hazard ratio.

80. loss of productivity **indirect cost**

Costs

There are three main types of costs in economic evaluation studies:

- **Direct** - those associated directly with the healthcare intervention (e.g. staff time, medical supplies, cost of travel for the patient, childcare costs for the patient, costs falling on other social sectors such as domestic help from social services)
- **Indirect** - those incurred by the reduced productivity of the patient (e.g. time of work, reduced work productivity, time spent caring for the patient by relatives)
- **Intangible** - those that are difficult to measure (e.g. pain or suffering on the part of the patient)

81. **less valid outliers**

82. **Best way of ITT per protocol analysis complete study growth curve**

[SPMM] 100 patients are enrolled in an RCT comparing an herbal medicine with fluoxetine at 12 weeks for depression. Out of 50 who were assigned to take fluoxetine, 8 did not turn up for follow-up. Out of 50 who were assigned to herbal treatment 12 did not complete the treatment or missing. If such a trial is carried out to see the biological efficacy of the new herbal medicine and not to compare its clinical effectiveness against fluoxetine, which of the following methods of analysis will be most suitable?
Select one:

- A. Sensitivity analysis
- B. Mean imputation method
- C. Last observation carried forward analysis

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D. Intention to treat (ITT) analysis

E. Per-protocol analysis

ITT may not suit those who are investigating biological effects of a drug via RCT model - those who do not take the drug cannot have a biological effect. But for those studies that report on effectiveness, ITT is invaluable. Using per-protocol (PP) analysis only patients who sufficiently complied with the trial's protocol are considered in the final analysis. This is the antithesis of intention to treat analysis. It is also called 'per-protocol treatment' analysis. PP analysis is often carried out in biological explanatory RCTs and as the secondary analysis in effectiveness trials.

[SPMM] In randomized controlled trials, intention-to-treat (ITT) analysis is used to
Select one:

A. Assess treatment effects as randomised despite a patient's nonadherence

B. Exclude subjects who were misallocated to an intervention

C. Prevent type 2 error in a trial

D. Reduce premature dropouts from occurring

E. Utilise at least 80% of the full analysis set

The intention-to-treat (ITT) analysis is one where every patient randomised to the clinical study enters the primary analysis. Any patients who drop out prematurely, are mis-allocated, or noncompliant, are also included in the primary analysis within the "as randomised" treatment group at allocation. As a result, the original randomisation is preserved and the potential bias due to exclusion of patients is avoided. Otherwise, if there are a substantial amount of non-random missing cases, then doing per-protocol analysis (excluding the non-compliant, missing cases) will increase type 1 error rate substantially. In ITT, the term "full analysis set" refers to the list of patients whose data is used for the primary analysis. It generally overlaps fully with the randomised set.

83. Consort venlafaxine & aripiprazole vs venlafaxine & placebo

See Mock 2 Q. 16-20

84. Respond venlafaxine alone EER

See Mock 2 Q. 16-20

85. ITT after continuation phase remission in arm1 compared arm2

See Mock 2 Q. 16-20

86. Most efficacious, tolerated **high left TMS**

See Mock 13 Q76

87. Most precise **priming**.

See Mock 13 Q76

88. Forest plot mirtazapine, placebo after SNRI

See Q45 Fresh questions 22

89. Difference in mean between 2 groups **BDI 1.7**

See Q45 Fresh questions 22

90. **pregabalin, gabapentin forest plot, Pregabalin more statistical in all**

<https://link.springer.com/article/10.1007/s40122-022-00451-4>

91. Gabapentin statistically less in the last 2 options

https://www.researchgate.net/figure/The-effect-of-pregabalin-illustrated-by-a-forest-plot-diagram-with-24-h-VAS-on-movement_fig6_315633907

92. Risk ratio can't calculate

ChatGPT Risk ratio, also known as relative risk, cannot be calculated when one of the groups being compared has zero events (e.g., zero cases of a disease or outcome). This situation arises in studies where there are no events in either the exposed or unexposed group, resulting in a division by zero error when attempting to calculate the risk ratio.

In such cases, alternative measures of association, such as risk difference or odds ratio, may be used instead, depending on the specific circumstances of the study and the available data. These alternative measures can still provide useful insights into the relationship between exposure and outcome when the risk ratio cannot be calculated.

- ➔ **Case control** = odds ratio can be calculated
- ➔ **Cohort** = calculate Relative risk RR

93. Odds ratio Case respond

94. SEM sample size 12 SD 16

$$\text{SEM} = \text{SD} / \sqrt{n} = 16 / \sqrt{12} = 4.624$$

95. SD ?? variance 36 6

$$\text{SD} = \sqrt{36} = 6$$

96. true about SEM

[SPMM] The term standard error of mean refers to
Select one:

- A. Dispersion of sample data around sample mean
- B. Error in computing the standard deviation
- C. Error in computing the sample mean
- D. Variance within the sample
- E. Dispersion of sample means around true mean**

The difference between the means of two samples, A and B, both randomly drawn from the same normally distributed source population, belongs to a normally distributed sampling distribution whose overall mean is equal to zero and whose standard deviation is given by the standard error.

97. Sensitivity 60%, Specificity 80%. Calculate LR- 1/2 0.5

- ➔ **LR+** = Sensitivity / 1- Specificity
- ➔ **LR-** = 1- sensitivity / Specificity

98. NNT 144

$$\text{NNT} = 1/\text{ABI (ARR)}$$

99. NNH 10 EER .5 CER .4 ABI 0.1

ABI (absolute benefit increase) = EER – CER = 0.5 – 0.4 = 0.1
NNH = 1/ABI = 1/0.1 = 10

100. Sample decreased CAM 23 to 11 but statistically non-significant what is needed ?

- A. Another study with large sample.
- B. Increase number of people until become significant.
- C. Clinically significant start ttt immediately.

ChatGPT If a study found a decrease from 23 to 11 in a sample size, but it was statistically non-significant, it suggests that the observed change may not be significant enough to confidently conclude that there is a true difference between the groups being compared. In such cases, increasing the sample size might help to detect smaller effects, making the study more statistically powerful.

However, it's important to consider other factors as well. Simply increasing the sample size may not always lead to statistical significance if the effect size is small or if there are other limitations in the study design or execution.

Furthermore, while statistical significance is important for drawing conclusions about the relationship between variables, clinical significance is also crucial. If the observed decrease in CAM score is not statistically significant but is clinically meaningful (i.e., it has practical implications for patient care), then immediate treatment might still be warranted based on clinical judgment and the overall context of the situation.

In summary, to address the situation:

- ➔ Consider whether increasing the sample size is feasible and whether it would likely lead to statistically significant results.
- ➔ Evaluate the clinical significance of the observed change and consider whether immediate treatment is warranted based on clinical judgment and patient needs.
- ➔ Assess other factors that may influence the interpretation of the results, such as study design, effect size, and potential confounders.

101. retrospective Cohort subject schiz and risk of cannabis ,pt with Depression risk of trauma

See Mock 1 Q21

	First-episode psychosis group (n=410)	Control group (n=370)	p value
Total population			
Lifetime history of cannabis use	..	..	0.277
Yes	275 (67%)	232 (63%)	..
No (never used)	135 (33%)	138 (37%)	..
Frequency of use	..	..	<0.0001
Less than once per week	68 (17%)	128 (35%)	..
At weekends	84 (20%)	63 (17%)	..
Every day	123 (30%)	41 (11%)	..
Most used type of cannabis	..	..	<0.0001
Never used	135 (33%)	138 (37%)	..
Hash-like	57 (14%)	162 (44%)	..
Skunk-like	218 (53%)	70 (19%)	..
Cannabis users			
Duration of use (years)	9.7 (7.4)	9.1 (7.8)	0.635
No details	3	1	..
Age at first cannabis use (years)	16.1 (4.2)	16.6 (3.2)	0.146
No details	3	1	..
Age at first use ≤15 years	..	..	0.028
No	172 (63%)	178 (77%)	..
Yes	100 (36%)	53 (23%)	..
No details	3	1	..

Data are n (%) or mean (SD) unless stated otherwise.

Table 2: Cannabis use

102. Qualitative study **exploration**

[SPMM] In which of the following studies a qualitative approach is most appropriate?

Select one:

- A. A comparison of the effectiveness of tetrabenazine vs. reserpine in the treatment of tardive dyskinesia.
- B. A survey of number of school teachers exposed to bullying from students and the distribution of mental health problems among those teachers.
- C. A multicenter pragmatic naturalistic exploration of all cause discontinuation rates for antidepressants.
- D. An exploration of the role of the 'Community team coordinators' in the elderly community health care teams in the North East of England.**
- E. A national postal survey of doctor's knowledge of prescribing nicotine replacement therapies.

Tetrabenazine trial is clearly quantitative. National surveys of knowledge and practice are usually conducted through questionnaires which collect feeble qualitative information; these are analysed quantitatively on most occasions. The last example is a pragmatic RCT, far from a qualitative study. Some situations where qualitative research is useful are,

1. When exploring the nature of the problem to be studied; i.e. before planning quantitative research
2. To generate hypothesis based on observations made.
3. Investigating anomalies or oddities observed during clinical practice
4. Examining policy implementation
5. Collating service users/ carers/ staff views.

103. Qualitative to quantitative **log transformation**

[SPMM] **Transformation of data:**

Parametric statistical tests offer more robust analysis than non-parametric tests. Hence, one may convert data into a form more suitable for parametric analysis using transformation methods. The most commonly used method is **log transformation**. For a right skew (positive) this yields a normal distribution (called lognormal curve). Such transformation can also linearize a nonlinear correlation with the upward shift. Square root transformation has normalizing and linearizing properties; in addition it can stabilize variance. It is used for normalizing Poisson distributions. Other types include reciprocal transformation, square transformation, and logit transformation. The reciprocal transformation is often used for analysis of survival rates. The logit transformation is often used for distribution of proportions. It linearizes a sigmoid curve. Note that confidence intervals may be difficult to interpret after transformation. Geometric mean values may get distorted in the process to some extent

104. Quality of reporting a systematic review of Meta analysis **Quorum**, PRISMA

[SPMM] Which of the following is an agreed method of assessing quality of conducting and reporting systematic reviews and metaanalyses?

Select one:

- A. ASSERT
- B. NICE
- C. SIGN
- D. CONSORT
- E. QUORUM**

The Quality of Reporting of Meta-analyses (QUOROM) conference was convened to address standards for improving the quality of reporting of meta-analyses of clinical randomised controlled trials (RCTs).

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The QUOROM group consisted of 30 clinical epidemiologists, clinicians, statisticians, editors, and researchers. In conference, the group was asked to identify items they thought should be included in a checklist of standards. A modified Delphi technique was used in assessing candidate items. The conference resulted in the QUOROM statement, a checklist, and a flow diagram.

You must be familiar with the following standards of publication:-

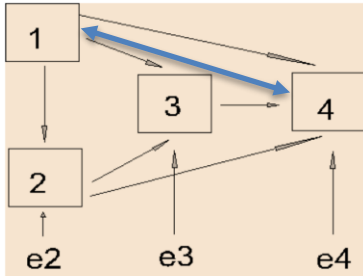
Guideline	Topic of guidelines
CONSORT (Consolidated Standards of Reporting Trials)	Randomised controlled trials
TREND (Transparent Reporting of Evaluations with Non-randomized Designs)	Non-randomised controlled trials
PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)	Systematic Reviews and Meta-Analyses (Replaced QUORUM - Quality of Reporting of Meta-Analyses)
MOOSE (Meta-analysis Of Observational Studies in Epidemiology)	Observational Studies in Epidemiology
STROBE (Strengthening the Reporting of Observational Studies in Epidemiology)	Observational Studies in Epidemiology
SQUIRE (Standards for Quality Improvement Reporting Excellence)	Quality improvement studies
STARD (Standards for the Reporting of Diagnostic accuracy studies)	Diagnostic studies
MIAME (Minimum Information about a Microarray Experiment)	Microarray studies
COREQ (Consolidated criteria for reporting qualitative research)	Qualitative studies

105. Path analysis causation shown by:

- A. Ext validity ,
- B. Int validity,
- C. P value,**
- D. Analogy

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Path analysis: This refers to causal modelling and prediction beyond simple regression. A set of independent variables is related to a set of dependent variables in these analyses. It can be used to examine situations in which there are various "chains" of influence among the variable studied. The independent (X) variables are called **exogenous variables** whereas the dependent (Y) variables are called **endogenous variables**. A path relationship is displayed using arrows that display presumed causal relations. A single-headed arrow flows from a putative cause to the effect. A double-headed, curved arrow indicates mere correlation but no predictive, causal links. A **path coefficient** indicates the direct effect of a variable assumed to be a cause on another variable assumed to be an effect. Path coefficients are written with two subscripts. The path from 1 to 2 is written p_{21} , the path to 2 from 1. Note that the effect is listed first. A path analysis in which the causal flow is unidirectional (no loops or reciprocal causes) is called **recursive**.



E = error

Double headed arrow = correlation but no causation

Points to note:

1. All possible paths from earlier to later variables are included in this particular graph (1 to 2, 3, & 4; 2 to 3, 4, 3 to 4). There are no backwards paths (e.g., 4 to 1).
2. The only exogenous variable is 1 (it has no arrows pointing to it).
3. There are 3 endogenous variables here (2,3,4). Each endogenous variable is explained by 1 or more variables in the model, plus an error term (e2 - e4). An endogenous variable can be a cause of another endogenous variable, but not for an exogenous variable.

(from MT Brannick; <http://faculty.cas.usf.edu/mbrannick/regression/Pathan.html>)

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106. Dichotomous Marital status

Dichotomous = yes/no questions

107. Difference between dichotomous data and another type:

- A. Dichotomous easier to represent,
- B. Easier to interpret,
- C. Have different way of testing

108. Categorical was:

- A. Age,
- B. Height,
- C. Gender,**
- D. Weight

109. Q about numbers 2*2 which test **Chi square**

EMI

110. EMI case hyperactive ADHD **methylphenidate**

Dexamphetamine
Methylphenidate
Atomoxetine
Clonidine
Amitriptyline
Methylphenidate sustained release
Advise parents and teachers as to the nature of handicap
Behavioural Therapy
Social skills training
Parent management training
Family therapy

ADHD treatment options

For each of the clinical situations related to ADHD described below, identify the correct mode of treatments from the list below

1. A 9-year-old school boy who needs once daily dosing (TWO options)
2. A 7-year-old boy has been diagnosed with ADHD. His father is keen to know the name of the drug that is usually the first line treatment for ADHD and is also well tolerated
3. A 7-year-old boy has been treated for ADHD for more than two years. Of late he grimaces and makes funny movements with his arms and shoulders, which is embarrassing for his parents (Choose TWO)
4. A 9-year-old girl requires treatment for ADHD symptoms. Of late she is also low in mood, not mixing with peer group, can't sleep and has no appetite
5. An 8-year-old boy has been treated for ADHD. Of late, he is becoming aggressive at home and school

Explanation:

- 1) **Methylphenidate** is now available in XL preparation. It could be administered once a day, preferably in the mornings. Atomoxetine can be given once daily and can be given at any time of day.
- 2) **Methylphenidate** is the first line treatment for ADHD, which is well tolerated and specifically treats ADHD core symptoms.
- 3) This boy has developed tics along with ADHD, possibly as a side effect of his medications. Although a trial of atomoxetine if tics worsen with stimulants, cases of tics with atomoxetine have been reported. Therefore, clonidine would be a good choice.
- 4) This girl is suffering from depression and ADHD. Tricyclics are helpful to treat both ADHD and comorbid depression/anxiety.
- 5) For children with ADHD on methylphenidate and has aggressive behaviour, Atomoxetine is unlikely to be beneficial. Consider behavioural therapy before treating them on antipsychotics.

111. **EMI another case Child**

Bipolar

112. EMI case Children 13 years mixed affective now depression on olanzapine 20mg

A. Fluoxetine

- B. Quetiapine
- C. Lurasidone
- D. Na li
- E. Carbamazepine
- F. Lamotrigine

[SPMM] A known bipolar patient has had many episodes of mania but no clinically significant depressive symptoms in the past. Currently, he presents with severe depression. He is already on lithium with good serum levels and is compliant. Of the listed options, what should be the next step?

Select one:

A. Add lamotrigine

- B. No intervention required
- C. Add venlafaxine
- D. Increase lithium dose
- E. Add sertraline

Explanation: As reflected in the most recent NICE, BAP, and CANMAT/ISBD bipolar treatment guidelines, there are several options in this scenario, including the addition of **olanzapine + fluoxetine, quetiapine, olanzapine, lamotrigine, or lurasidone.**

1. **Fluoxetine + Olanzapine**
2. **Quetiapine only**
3. **Olanzapine only**
4. **Lamotrigine only**

[SPMM] Which of the following should be used as a first choice treatment in adolescents with acute manic or mixed episodes?

Select one:

A. Olanzapine

- B. Lithium

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- C. Carbamazepine
- D. Sodium valproate
- E. Lamotrigine

Explanation: Aripiprazole and lithium are the only licensed medications for treating mania. In practice, lithium is not generally used acutely in the treatment of manic/mixed episodes in adolescents, as it requires commitment to longer-term treatment and regular blood tests. NICE guidelines (updated 2020) recommend considering **aripiprazole** or using an alternative antipsychotic (e.g. **risperidone, olanzapine, quetiapine**) in accordance with guidance for adults and limiting drug treatment to 12 weeks. It is unclear whether valproate has the same beneficial effects on mania in young people as it does in adults. NICE recommends that valproate monotherapy should NOT be used for first-line management of mania and it should NOT be offered to girls or young women of childbearing potential because of teratogenic effects and the risk of polycystic ovary syndrome

Treatment for mania or hypomania

NICE recommends the treatments described in this section for adults aged 18 and over. Young people aged 13 and over may be offered a medication called aripiprazole (see other NICE guidance for details of our guidance on aripiprazole), or they may be offered one of the treatments for adults. Young people should not usually take antipsychotic medication for more than 3 months.

Re: Aripiprazole for mania in Adolescents "The Appraisal Committee reviewed the data available on the clinical and cost effectiveness of aripiprazole, having considered evidence on the nature of acute manic episodes in bipolar I disorder in adolescents and the value placed on the benefits of aripiprazole by people with the condition, and clinical specialists. It also took into account the effective use of NHS resources"

113. EMI case Pregnant 1st trimester bipolar depression first time **Quetiapine**

[SPMM] A patient with bipolar disorder and a history of mixed affective states is in the first trimester of her pregnancy. She has not been on any medications of late, but is very concerned about the possibility of relapse. Of the following, what would be the preferred medication to prescribe to her?

Select one:

- A. Olanzapine**
- B. Valproate and lithium
- C. Lithium
- D. Valproate
- E. Lamotrigine

Explanation: For pregnant women with bipolar disorder who have had a long period without relapse, the possibility of switching to a safer drug (antipsychotic) or withdrawing treatment completely before conception and for at least the first trimester should be considered. No mood stabiliser is clearly safe. NICE recommends the use of mood stabilising antipsychotics as a preferable alternative to a mood stabiliser. **ECT can be considered in pregnancy**

[SPMM] In the UK, which of the following is the most commonly used antipsychotic in pregnancy?

Select one:

- A. Olanzapine
- B. Amisulpride
- C. Quetiapine**
- D. Risperidone
- E. Aripiprazole

Explanation: In contrast to previous decades, quetiapine (and then aripiprazole) has become more commonly used during pregnancy than olanzapine in the UK. Most reproductive safety data are available for quetiapine, olanzapine, risperidone, and haloperidol, with more limited data for clozapine, aripiprazole, and ziprasidone. Quetiapine has a relatively low rate of placental passage.

- ➔ **Quetiapine** is safe in pregnancy as It has relatively low rate of crossing placenta
- ➔ **Olanzapine, quetiapine and risperidone** are safe in breastfeeding

114. EMI case Bipolar man mixed affective **Olanzapine**

[SPMM] Mr Evans is a 26-year-old man who has been brought to his local A&E by his family after becoming increasingly elated and energetic over the last week. He stopped sleeping and started to build a "perpetual motion machine" in his garage. Which of the following would be recommended to treat him?

Select one:

- A. Lithium
- B. Aripiprazole
- C. Carbamazepine
- D. Olanzapine**
- E. Sodium Valproate

NICE Bipolar guideline recommends that individuals with a manic episode should be offered **haloperidol, olanzapine, quetiapine or risperidone** in the first instance, depending upon patient preference, advance directives and clinical context.

Dementia types

115. EMI Visual hallucination selegiline 3 years ago **Parkinson dementia**

Dementia- subtypes
For each of the following descriptions choose the most appropriate diagnosis

A 60-year-old woman has been wandering the streets and was taken to A&E. She appears unconcerned during the assessment. She has diabetes and hypertension. CT shows a subcortical pattern of excessive white matter lesions with no other notable findings.

A 70-year-old man with a history of recent onset cognitive decline and visual hallucinations. He has been attending the neurology clinic for the past three years, and his medication includes a list of drugs also to Selegiline.

A 67-year-old man with a history of progressive cognitive decline, loss of visuospatial and calculation abilities and reduction in functioning.

Choose...

- Normal pressure hydrocephalus
- Depressive pseudodementia
- Alcoholic dementia
- Dementia in Pick's disease
- Wilson's' disease.
- Dementia in Huntington's disease
- Lewy body Dementia
- Binswanger's disease
- Parkinson's disease dementia
- Sporadic CJD
- Alzheimer's disease

Explanation:

1) Case 1 is **Binswanger's disease** (a progressive small vessel disease). This is subcortical dementia with a clinical course characterized by slow intellectual decline and generalised slowing. There is a history of vascular risk factors like diabetes and hypertension. The clinical picture may be dominated by dementia, and there may be concomitant physical problems such as gait disorders or dysarthria. MRI scan shows small distinct infarcts (lacunae) or more generalised white matter changes (Leukoariosis).

2) This patient is on Selegiline, which is a drug used for Parkinson's disease and has been reviewed by the neurology team. It is a case of **Parkinson's disease dementia**. If the **parkinsonian symptoms have existed for more than 12 months before dementia develops** then a diagnosis of Parkinson's disease dementia is given. Otherwise, you can consider Lewy Body Dementia as an option.

3) It is a case of **Alzheimer's** dementia featured a history of progressive cognitive decline and reduction in functioning. It is a type of cortical dementia involving temporoparietal dysfunction such as loss of visuospatial and calculation abilities and reduction in functioning

116. EMI **Stomping** gait, unexplained falls, dysarthria grandfather MI **Fredrick ataxia**

[SPMM] A 30-year-old man with clumsy gait, repeated falls and visual problems reports that his grandfather reportedly developed the same condition and died at an age of 41 after myocardial infarction. The most likely diagnosis to be considered is

Select one:

- A. Wilson's disease
- B. MELAS syndrome
- C. Fragile X syndrome
- D. Myotonic dystrophy
- E. Friedreich's ataxia**

In Friedreich's ataxia the spinal and peripheral nerves and cerebellum degenerate, affecting balance and movement and sensory functions but with intact cognition. The disorder also causes problems in the heart and spine, and some people with the condition develop diabetes. Various forms of heart disease often accompany Friedreich's ataxia, including hypertrophic cardiomyopathy, myocardial fibrosis, and cardiac failure. Heart block is also common. Heart disease is the most common cause of death. Friedreich's ataxia is an autosomal recessive, trinucleotide expansion disease affecting GAA sequence of frataxin protein encoding gene on chromosome 9.

Gait	Description	Associated conditions
Festinating (Shuffling)	Posture is stooped forward. Gait initiation is slow and steps are small and shuffling	Parkinson's disease
Ataxic	Gait is wide-based with truncal instability and irregular lurching steps which results in lateral veering and if severe, falling	Cerebellar disease, e.g. Wernicke's
Antalgic	Stance phase of gait is abnormally shortened relative to the swing phase, usually done to minimise pain	Lower limb trauma
Spastic (scissor or diplegic)	Rigidity and excessive adduction of the leg in swing, plantar flexion of the ankle, flexion at the knee, adduction and internal rotation at the hip, and contractures of all spastic muscles	Cerebral palsy
Steppage (neuropathic or equine)	High-stepping gait so as to prevent scraping of the toe on the ground	Foot drop
Myopathic (waddling)	A broad-based gait with a duck-like waddle to the swing phase, the pelvis drops to the side of the leg being raised with forward curvature of the lumbar spine, and a marked body swing	Proximal myopathy
Pigeon	Toe(s) point(s) inwards when walking	Structural abnormalities
Trendelenburg	During the stance phase, the weakened abductor muscles allow the pelvis to tilt down on the opposite side. To compensate, the trunk lurches to the weakened side to attempt to maintain a level pelvis throughout the gait cycle	Poliomyelitis or muscular dystrophy
Stomping	Bilateral high steppage due to lack of proprioception	Friedreich's ataxia
Magnetic	Feet seem as if magnetically attracted to the floor	Normal pressure hydrocephalus
Choreiform (hyperkinetic)	Irregular, jerky, involuntary movements in all extremities. Walking may accentuate their baseline movement disorder	Sydenham chorea, Huntington's
Sensory	In an effort to know when the feet land and its location, the patient will slam the foot hard onto the ground in order to sense it	proprioceptive loss e.g. diabetes, B12 deficiency

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117. EMI unconcern **binswanger dementia** (See above)

118. EMI dementia types 70y cubes pentagon **AD**

[SPMM] Mrs Y is a 73-year-old woman who has short-term memory loss for last eight months. On MMSE, she has problems with calculation and could not draw a pentagon on MMSE. She is not working. Her mood is described as euthymic.

2) It is a case of dementia featured by a history of progressive cognitive decline and reduction in functioning. It is a type of cortical dementia involving temporo-parietal dysfunction such as loss of visuospatial and calculation abilities and reduction in functioning.

119. EMI pt who had renal transplantation and complaint of recent memory loss imaging show frontal lobe lesions **Progressive multifocal leukoencephalopathy** FTD

Renal transplant = blood transfusion and immunocompromised

[SPMM] Kieran Todd is a 53-year-old man of Kenyan origin who presents to his GP with gradual onset and progressive head tremor, difficulties with coordination and pronouncing of some words, and behavioural changes (eg increased impulsivity and emotional incongruence). His GP does a routine set of bloods, and discovers Mr Todd is HIV positive. He refers Mr Todd for an MRI of the brain. What is the most likely brain change associated with HIV?

Select one:

- A. Hydrocephalus
- B. Reversal of asymmetry of planum temporale
- C. Progressive multifocal leukoencephalopathy**
- D. Symmetrical hyperintensity of the pulvinar nuclei of the thalamus
- E. Atrophy of angular gyrus

Progressive multifocal leukoencephalopathy (PML) is associated with HIV; HIV infection accounts for almost 85% of the total cases of PML. It is currently one of the AIDS-defining illnesses in HIV-infected patients

Family therapy

120. EMI Paradoxical injunction **systemic** strategic

121. EMI Unspoken rules **structural**

122. EMI Past experiences **psychodynamic**

Approaches to psychotherapy in youth mental health
From the list provided, choose the type of family therapy that is characterised by the description.

1. Paradoxical interventions are commonly used ✗
2. Introduced the concepts of 're-framing' the problem and hypothesising, as well as ideas about neutrality and curiosity; therapist takes more of a non-expert stance ✗
3. Challenges to the hierarchy within the family results in problems; therapist takes a directive, 'expert' stance ✗
4. Difficulties are a reflection of dominant, problem-saturated stories; therapist highlights 'unique outcomes' and 'externalisation' ✗
5. Current family functioning is grounded in the past; core concepts include splitting; therapist creates a holding environment and offers interpretations ✗
6. Clear rules govern optimal family organisation, with a focus on hierarchy, subsystems, and boundaries ✗

In the model of **strategic family therapy**, problems always arise because of difficulties with hierarchy within the family system. Rather than attempting to resolve this, the family is ambivalent about having the problem, as it provides some gain for them. Reflecting this idea, the In Milan systemic family therapy, there is an emphasis on family beliefs and meanings, and the idea of there being no single objective truth about the problem.

The **Milan model** introduced the concepts of 're-framing' the problem and hypothesising, as well as ideas about neutrality and curiosity, with the therapist taking more of a 'not-knowing', non-expert stance.

In **structural family therapy**, Minuchin proposed that clear rules govern optimal family organisation and structure, with a focus on hierarchy, subsystems, and boundaries. Challenges to this structure result in problems which the family attempts to address. In this model, the therapist takes a directive, 'expert' stance.

In **narrative family therapy**, difficulties are a reflection of unhelpful, dominant, 'problem-saturated' narratives. The therapist helps to highlight 'unique outcomes' to challenge this narrative and, through the use of 'externalisation' (i.e. separating the problem from the person), helps to rewrite this to a more helpful one.

In **psychodynamic-psychoanalytic family therapy**, core concepts include object relations, projective identification, splitting, and scapegoating. Common strategies include creation of a holding environment, interpretation, and emphasis on therapeutic alliance. This type of family therapy probably influences family therapists' clinical work more than is usually acknowledged. therapist takes a strategic stance to overcome their resistance, such as using 'paradox' (e.g. suggesting the problem may not be resolvable).

- **Paradoxical Injunction → Systemic family therapy**
- **Intervention paradoxical → Strategic family therapy**

ECG

123. EMI Anorexia restriction food binge eating hypokalemia **Uwave prolonged PR**
See Q.94 Fresh Q. 2022
124. EMI Clozapine **saddle T inversion**
See Q.94 Fresh Q. 2022
125. EMI Citalopram 20mg,diphenhydramine **QTC prolongation**
See Q.94 Fresh Q. 2022
126. 125-EMI Donepezil 10 mg **Sinus rhythm with decreased rate**
127. **See Q.94 Fresh Q. 2022**

Endocrine

128. EMI low mood obesity in face neck abdomen upper back
- A. **Cushing,**
 - B. Addison,
 - C. hypothyroidism,
 - D. hypoparathyroidism,
 - E. Hypopituitarism,

- F. Folate deficiency,
- G. Vit B

129. EMI depression h/o intake of phenytoin for epilepsy, with tongue red, ulcer on mouth decreased sense of taste ? **Folate deficiency**
130. EMI memory loss, tingling numbness, tongue change, Low mood, tingling in hands and feet, double vision, muscle cramps and weakness, pale yellow skin, tongue is smooth and ? **Vit B**
131. EMI 15 year old with low mood, emotional dysregulation, dry skin, nails Numbness and tingling of lips, coarse dry skin, ocular cataracts, trophic changes in nails ? **hypoparathyroidism**

Treatment options of reversible dementia (2)	
For each clinical situation given below, which of the listed treatments may result in a partial or fully recoverable state?	
A 45-year-old chronic alcoholic now presents with apathy, fatigue, irritability, depression and cognitive impairment.	Choose...
A 76-year-old woman with atrophic gastritis now presents with fatigue, tingling and numbness of hands & feet, depression and memory problems.	Choose...
A 57-year-old woman presents with mouth ulcers, diarrhoea, lassitude, impaired cognition, depression and dementia	Choose...
A 45-year-old woman with lethargy, excessive tiredness, constipation, low mood and memory problems.	Choose...

Explanation:

- 1) The diagnosis is **chronic thiamine deficiency** due to alcoholism. Acute thiamine depletion causes Wernicke's encephalopathy and later, Korsakoff's psychosis. Chronic thiamine deficiency causes apathy, fatigue, irritability, depression and cognitive impairment prior to the development of classical beriberi.
- 2) The diagnosis is **vitamin B 12 deficiency** (Pernicious anemia). The typical symptoms of vitamin B12 deficiency anemia include diarrhoea or constipation, fatigue, loss of appetite, pale skin, problems concentrating, swollen, red tongue or bleeding gums. Monthly vitamin B12 injections are prescribed to correct the vitamin B12 deficiency.
- 3) The diagnosis is **folate deficiency**. It may cause diarrhea, grey hair, mouth ulcers and peptic ulcers. It may be associated with lassitude, impaired cognition, depression and dementia (prolonged deficiency). The treatment would be replacement with folic acid supplements and fortified foods, to ensure the proper daily intake.
- 4) This is **hypothyroidism**, caused by low thyroxine levels due to thyroid disease, thyroidectomy or drugs. Mental symptoms develop insidiously and may include impaired concentration, psychomotor retardation, depression and dementia and 'myxedematous madness', a form of paranoid psychosis with delusions and hallucinations. The psychiatric symptoms usually resolve with thyroxine replacement Therapy.

[SPMM] Following thyroid surgery, a 45-year-old woman became depressed and irritable. She was observed to have twitchy movements in her face:

This patient has developed features of **hypoparathyroidism** following thyroid surgery. The common features include depression, anxiety, irritability, acute organic reaction and cognitive impairment. The psychiatric symptoms increase as hypocalcemia worsens but they respond rapidly to treatment with vitamin D. The physical symptoms include tetany, perioral anesthesia, carpopedal spasm and tapping over parotid causes facial muscles to twitch (Chvostek's sign).

[SPMM] A 27-year-old woman has weight gain and hirsutism. She is also clinically depressed and has a history of hypertension.

Cushing's syndrome: Clinical features include weight gain, menstrual irregularity, amenorrhoea, hirsutism in females, impotence in males, depression, muscle weakness and fractures. Signs include tissue wasting, myopathy, thin skin, purple abdominal striae, osteoporosis, hirsutism, hypertension, hyperglycemia, predisposition to infection and poor wound healing.

LD EMI

132. EMI LD complaint butterflies on your stomach

- A. Diagnostic overshadowing,
- B. Intellectual distortion,
- C. Cognitive disintegration,
- D. Psychosocial normalization

[SPMM] Mr Jason Hughes is a 42-year-old man with a diagnosis of Down's Syndrome and mild intellectual disability. He largely lives independently with some community support and works for a local charity. He and his family feel it is important that he has a social role and life circumstances that are not too different from others of his age. Which of the following best describes this social phenomenon?

Select one:

- A. Deinstitutionalisation
- B. Assimilation
- C. Normalisation**
- D. Acclimatisation
- E. Social role valorisation

Normalisation refers to the principle that people with intellectual disabilities should be able to experience 'normal patterns' of everyday life, such as places of accommodation, and the range of day-to-day activities. Rather than forcing individuals to conform to societal norms, the principle argues for changing the environment, not the individual, to provide the required normalcy.

[SPMM] Miss Olive Charleston is a 32-year-old woman with moderate intellectual disability and has been suffering from otitis externa for the past 10 days. The staff members in her care home have observed her banging her head on doors and tables. They attributed this behaviour to her intellectual disability and have been trying to reassure and distract her without success. Which of the following best describes this phenomenon?

Select one:

- A. Retrospective diagnosis
- B. Institutionalisation
- C. Diagnostic overshadowing**
- D. Stereotyping
- E. Neglect

Diagnostic overshadowing is the attribution of a person's symptoms to their mental condition, when such symptoms actually suggest a comorbid condition. The term was first used to describe the under-diagnosis of mental illness in people with intellectual disability. In recent years, the term has also been used when physical illnesses are overlooked in people with mental illness.

133. EMI LD bipolar told dr I learnt driving but no concern then pt said I have episode and dr driving one of sign the same options

Types of data

134. EMI: chi square **dichomitus**

[Psych-Mentor] Dichotomous (binary) outcome data arise when the outcome for every participant is one of two possibilities, for example, dead or alive, or clinical improvement or no clinical improvement. The most commonly encountered effect measures used in clinical trials with dichotomous data are:

1. The odds ratio (OR)
2. The risk ratio (RR) (also called the relative risk)
3. The risk difference (RD) (also called the absolute risk reduction)
4. The number needed to treat (NNT)

135. EMI scale from 1-5 one is mild and five is severe **ordinal**

Data Types
For each of the following descriptions choose an appropriate response from given list:
A prospective study investigates relapse rates of depression in patients who have maintenance ECT.
A neuroimaging study measures insight using a single item as good, moderate, mildly abnormal, poor or absent to correlate with parietal volume.
Lean body mass on admission at an eating disorder unit.

Choose...
Ratio
Multiple Nominal
Ordinal
Binary Nominal
Interval
Independent

Explanation: Having a relapse or not is a categorical outcome (i.e. 'yes' or 'no' type). Hence, the outcome is **nominal**.

Likert scales, as described here in the insight measurement, produce **ordinal** data.

Lean body mass is a continuous variable with a meaningful zero. So one can say that subject A is two times heavier than subject B. As such ratios are possible, lean body mass is a **ratio data**. Consider **interval** scales such as dates of admission to a psychiatric unit. These are ordered and have real numerical units and can be subtracted from each other, but there is no intrinsic zero to the scale so that dividing them does not make sense. For example, it would be absurd to say one date of admission is three times 'bigger' than another date.

Type of study

136. EMI Giving scenario outcome **Case control**137. EMI parents have ID and we want to follow up performance of their children at school **Cohort**138. EMI effectiveness of ECT **RCT**

Study Design 2
For each of the studies described below, select the most appropriate design from the list
To investigate the relationship between alcohol consumption and depression, a group of patients attending the psychiatric outpatient clinic with a diagnosis of depression were questioned about their alcohol consumption. A group of non-depressed age and sex matched patients attending the clinic were also questioned about their alcohol consumption using an identical protocol.
To investigate the relationship between certain solvents and dementia, all employees at a chemical firm were questioned about their exposure to an industrial solvent, and the amount and length of exposure measured. These subjects were regularly followed-up, and after 20 years a copy of the death certificate for all those who had died was obtained.
A survey was conducted of a random sample of junior doctors employed at a particular hospital. Among other questions, the questionnaire asked about the grade of the doctor and whether he/she was satisfied with the psychiatric training they were receiving.
A group of patients were invited to a focus group to elicit their views on drug treatment and reasons for non-compliance.
In unemployed adults with severe mental illnesses, the effectiveness of assertive community treatment (ACT) compared to usual care has been previously established at your local hospital. Clinical Director in your trust is interested in investigating if ACT is more expensive than the usual care.

Choose...
N-of-1 trial
Factorial RCT
Cross Sectional Study
Epidemiological study
Case-Control Study
Economic analysis
Meta-analysis
Randomized controlled trial
Cohort Study
Qualitative study
Choose...
Choose...

Explanation: 1) In cases (depressed patients) and controls (non-depressed individuals) are recruited simultaneously to study an association (depression and alcohol). Hence, this is a **case-control study**.

2) In a **cohort** of those who were exposed to the solvents at a factory is investigated in a cohort study.

3) In though the sample is random, two factors are evaluated simultaneously to study their co-occurrence. This is a **cross-sectional survey**.

4) Focuses on subjective experiences - it is a **qualitative study**.

5) Requires an **economic analysis** of the hospital data

Study designs III

Choose the most ethically appropriate and practically possible **study design** from the list for each research question below:

A very costly intervention is introduced as a control measure of multiple sclerosis though it does not cure it. A neurology team wants to compare this treatment with placebo, but they want to recruit as few participants as possible.

A clinical researcher wants to compare whether a new drug has fewer side effects than another established drug.

A nursing research team wishes to get patient's perspectives on the quality of care they receive as inpatients.

A researcher wants to evaluate four treatments for psychotic depression - antidepressants alone, antipsychotics alone, antidepressants and antipsychotics together, neither (placebo).

Choose...

Ecological survey
Clinical audit
Prospective Cohort
Case-control study
Crossover trial
Qualitative study
Open label RCT
Factorial RCT
Parallel RCT
Cross-sectional survey

Explanation:

An RCT has a cross-over design when each of the participants is given all the study interventions in successive periods, with the order being determined at random. Crossover trials produce within participant comparisons, whereas parallel designs produce between participant comparisons. As each participant is also a control in crossover trials, valid results can be obtained with fewer participants than would be required with a parallel design. Hence, this design is likely to be the most suited for Q1.

Please remember that most RCTs have a parallel design. In RCTs with parallel group design, each group of participants is exposed to only one of the study interventions. This is useful for comparing both beneficial and adverse effects of either an active drug against placebo (Q2) or two or more active drugs.

For Q3 where 'values' are measured, a qualitative design is appropriate.

When two or more experimental interventions are not only compared separately but also in combination and against a control in an RCT, then it is called a factorial RCT.

139. EMI percentage of P value **5** 0.5

140. EMI The point at which the confidence interval for a difference between means becomes non-significant

- A. **Zero**
- B. One

Type of chart

141. EMI. On choosing the best chart plot represent the data. (funnel, bar, frequency, curve scatterplot. Pie.)

Data

Which of the following ways of graphically displaying data would be most appropriate for each of the following circumstances?

1. Displaying the results of several different randomised control trials which have addressed the same research question.

2. A randomised controlled trial is conducted into a new antidepressant. The authors wish to compare the rate of response across 5 age groupings.

3. A meta-analysis is conducted into the efficacy of an antipsychotic medication. The authors are aware that the various studies into its use have led to significantly different results and wish to try and gain an overview of them to check for bias.

Bar chart
Histogram
Funnel plot
Forest plot
Pie chart

Explanation:

1) A forest plot, also known as a blobbogram, is a graphical display of estimated results from a number of scientific studies addressing the same question, along with the overall results.

2) Bar charts are used to show comparisons among discrete categories. One axis of the chart shows the specific categories being compared, for example, age groupings, and the other axis represents a measured value, such as rates of response to medication. In comparison, a histogram only shows the distribution of continuous numerical data, for example, the number of people in each age grouping.

3) This is most likely to refer to a funnel plot. This is a graph designed to check for the existence of publication bias. They are commonly used in systematic reviews and meta-analyses. In the absence of publication bias, it assumes that studies with high precision will be plotted near the average, and studies with low precision will be spread evenly on both sides of the average, creating a roughly funnel-shaped distribution.

Graph type	Summary
Box-and-whisker plot	Graphical representation of the sample minimum, lower quartile, median, upper quartile and sample maximum
Funnel plot	Used to demonstrate the existence of publication bias in meta-analyses
Histogram	A graphical display of continuous data that shows data by way of bars to display the frequency of numerical data
Bar Chart	A graphical display used to compare the frequency, total count, sum, or an average of data in different categories by using horizontal or vertical bars. It is also known as a column chart
Forest plot	Forest plots are usually found in meta-analyses and provide a graphical representation of the strength of evidence of the constituent trials
Scatter plot	Graphical representation using Cartesian coordinates to display values for two variables for a set of data

Graph type	Summary
Kaplan-Meier survival plot	A plot of the Kaplan-Meier estimate of the survival function showing decreasing survival with time

Data type	Description
Nominal	Observed values can be put into set categories which have no particular order or hierarchy. You can count but not order or measure nominal data (for example birthplace)
Ordinal	Observed values can be put into set categories which themselves can be ordered (for example NYHA classification of heart failure symptoms)
Discrete	Observed values are confined to a certain values, usually a finite number of whole numbers (for example the number of asthma exacerbations in a year)
Continuous	Data can take any value with certain range (for example weight)
Binomial	Data may take one of two values (for example gender)
Interval	A measurement where the difference between two values is meaningful, such that equal differences between values correspond to real differences between the quantities that the scale measures (for example temperature)
Ratio	A measurement where not only intervals but also ratios between numbers are meaningful due to a non-arbitrary zero point (for example weight, height)

Theme: Graphical representations of statistical data

- A. Forest plot
- B. Funnel plot
- C. Box plot
- D. Histogram
- E. Box-and-whisker plot
- F. Bar chart
- G. Stem plot
- H. Pearson survival plot
- I. Kaplan-Meier survival plot
- J. Scatter plot

Match each one of the following descriptions to the appropriate type of graph

68. A plot of the estimate of a patient's survival showing decreasing survival with time

CORRECT ANSWER : **Kaplan-Meier survival plot**

69. A graphical display of continuous data where the values have been categorised into a number of categories

CORRECT ANSWER : **Histogram**

70. Used to demonstrate the existence of publication bias in meta-analyses

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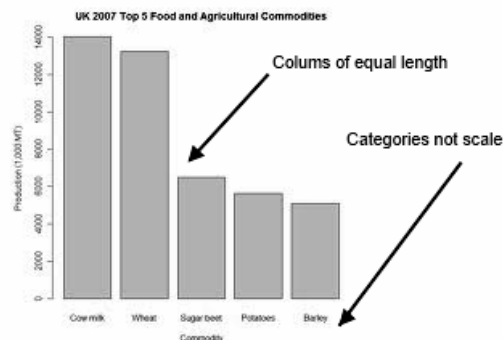
CORRECT ANSWER : **Funnel plot**

Regarding histograms, which of the following is true?

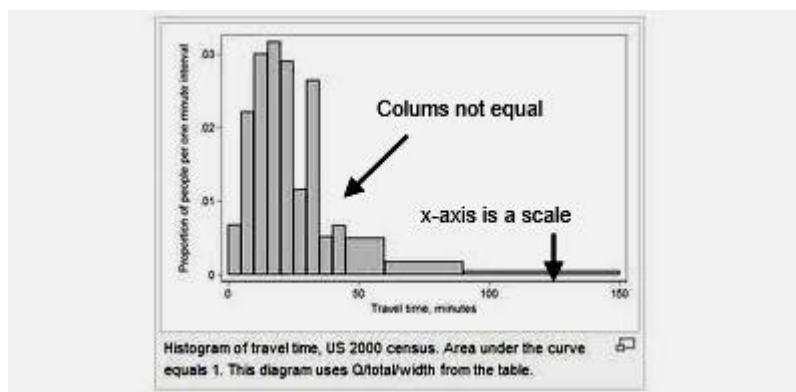
- A. The horizontal axis does not have a scale
- B. The quantity on the vertical axis is the frequency density**
- C. They are best used for categorical data (*continuous not categorical*)
- D. The height of the bar indicates the frequency
- E. Columns must have the same width

Stats Histograms

A bar chart is used to summarise ordinal or nominal data. Conventionally the x-axis represents the categories (and therefore does not have a scale) and the y-axis the frequencies. Columns are of equal length and the height of the bar indicates the frequency.



Histograms are similar to bar charts except that the width of the columns can be different. Histograms are used for quantitative data. In a histogram both axes have a scale. The value of the y-axis is the relative frequency (aka the frequency density). **The area (not height) of a column indicates the true frequency.**



142. representation of people with different age groups in a single area

- A. bar chart
- B. pie chart

Plots & Graphs (1)
Using the definitions given below, identify the type of plots/graphs used in statistical illustrations:

A survival curve in which the survival probability is plotted against the time from baseline.

A graph showing the median, range and interquartile range of a set of values

A diagram showing the proportional distribution of a categorical variable using a divided circle.

A diagram that illustrates the distribution of a categorical variable by showing a separate horizontal or vertical bar for each 'category', its length being proportional to the number of observations in that 'category'

Choose...

- Pie Chart
- Dot plot
- Kaplan-Meier plot
- Forest plot
- Bar chart
- Fagan's Nomogram
- Histogram
- Normal plot
- Scatter plot
- Box and whisker plot

Explanation:

- 1) Kaplan-Meier plot is a survival curve in which the survival probability is plotted against the time from baseline. It is used when exact times to reach the endpoint.
- 2) A box-and-whisker plot is a diagram illustrating the distribution of a variable; it indicates the median, upper and lower quartiles, and, often, the maximum and minimum values.
- 3) A pie chart is a diagram showing the frequency/proportional distribution of a categorical or discrete variable. A circular 'pie' is split into sections, one for each 'category'; the area of each section is proportional to the frequency in that category.
- 4) A bar or column chart is a diagram that illustrates the distribution of a categorical or discrete variable by showing a separate horizontal or vertical bar for each 'category', its length being proportional to the (relative) frequency in that 'category'.

Plots & Graphs (2)
Using the definitions given below, identify the type of plots/graphs used in statistical

A diagram relating the pre-test probability of a diagnostic test result to the likelihood and the post-test probability.

A diagram used in a meta-analysis showing the estimated effect of each trial and the pooled average with confidence intervals

A diagram that illustrates the relative frequency distribution of a continuous variable by using connected bars.

The two-dimensional plot of one variable against another, with each pair of observations marked by a point

A mixture of a diagram and a table used to illustrate the distribution of data.

Choose...

- Scatter plot
- Steam and leaf plot
- Histogram
- Forest plot
- Dot plot
- Bar chart
- Fagan's Nomogram
- Kaplan-Meier plot
- Box and whisker plot
- Pie Chart

Your answer is incorrect.

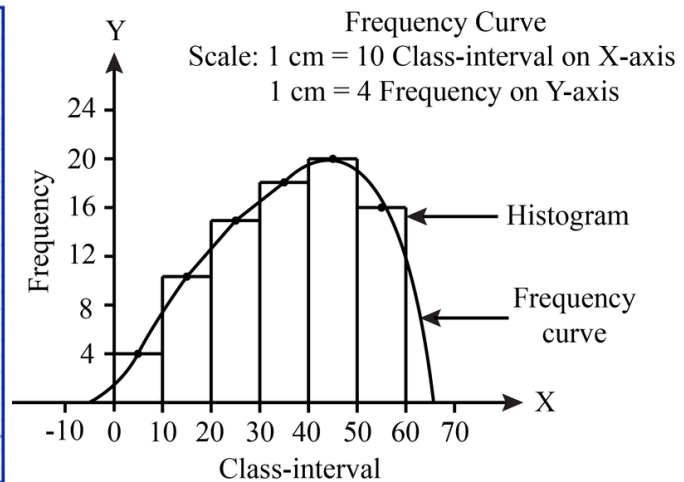
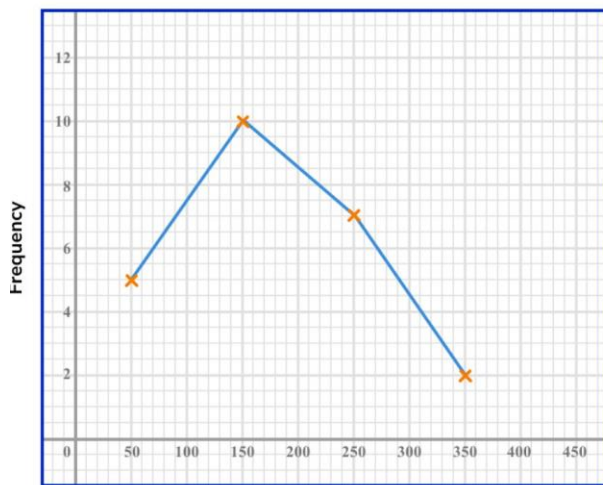
Explanation:

- 1) Fagan's Nomogram (also called Bayesian Nomogram) is a diagram relating the pre-test probability of a diagnostic test result to the likelihood and the post-test probability. It is usually used to convert the former into the latter.
- 2). A forest plot is a diagram used in a meta-analysis showing the estimated effect in each trial and their average (with confidence intervals)
- 3). A Histogram is a graph of continuous data with the data categorized into many classes. It illustrates the (relative) frequency distribution of a continuous variable by using connected bars. The bar's area is proportional to the (relative) frequency in the range specified by the boundaries of the bar.
- 4) Scatter diagram is a two-dimensional plot of one variable against another, with each pair of observations marked by a point
- 5). A steam and leaf plot is a mixture of a diagram and a table used to illustrate the distribution of data. It is similar to a histogram and is effectively the data values displayed in increasing order of size.

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143. Association relationship between cholesterol level and something else **scatter plot**

144. 3 categories in 4 wards presented by which graph? frequency curve



145. EMI validity suicide **predictive**

146. EMI test on autism concurrent construct content

[SPMM] The validity of a new instrument is compared with an external criterion. A conceptually related external criterion is identified to occur sometime in the future. If the correlation between current scores obtained using the instrument and the future expected outcome is studied, this is called
Select one:

- A. Incremental validity
- B. Predictive validity**
- C. Inter-rater reliability
- D. Concurrent validity
- E. Internal consistency

As something happening in future is assessed against the scale's current measures, predictive validity is being tested here

[SPMM] If MRCPsych Paper B accurately indicates participants' scores on MRCPsych CASC exams taken at a later date, then MRCPsych Paper B is said to have which of the following

Select one:

- A. Concurrent validity
- B. Retrospective validity
- C. Content validity
- D. Predictive validity**
- E. Face validity

In predictive validity, we assess the ability of a measurement scale to predict something it should theoretically be able to predict. For instance, we might theorize that a measure of math ability should be able to predict how well a person will do in an engineering-based profession. We could give our measure to experienced engineers and see if there is a high correlation between scores on the measure and their salaries as engineers. A high correlation would provide evidence for predictive validity -- it would show that our measure can correctly predict something that we theoretically think it should be able to predict.

Psychometrics of rating scales	
Which of the listed terms best describes the attributes of the scales that are being tested in each of the following scenarios?	
1. You are administering a putative test of cognition and MMSE (mini-mental state examination) at the same time, and the correlation between the two is computed	☒ Divergent validity ☐ Predictive validity ☐ Content validity ☐ Marketing validity ☐ Face validity ☐ Convergent validity ☐ Inter-rater reliability ☐ Concurrent validity ☐ Test re-test reliability ☐ External validity
2. A new self-report questionnaire for PTSD called NEO-PTSD has questions covering the three major dimensions of PTSD - hyperarousal symptoms, avoidance and intrusions - in a detailed manner.	☒
3. A medical student has noted that a specific scale has been repeatedly used in several controlled trials evaluating treatments for early onset schizophrenia. He is now keen to know the extent to which the observations using this scale can be generalized to a wider population of patients.	☒
4. The extent to which the same trait is measured by different methods	☒
5. The extent to which an instrument distinguishes the various distinct constructs that may relate to each other	☒
6. A research team wants to assess whether the acute stress disorder related dissociation in trauma-exposed youth accounts for later PTSD in adult life.	☒

Explanation: 1) **Concurrent validity** reflects the agreement of a new instrument with another scale, which has itself been shown to be valid. It is usually determined by comparing the new scale within an external gold standard.

2) **Content validity** refers to whether the contents i.e. each sub-scales, items or elements of the test are in line with the general objectives or specifications the test was originally designed to measure. A good coverage of all domains thought to be related to the measured condition would improve the content validity of an instrument.

3) **External validity** describes the extent to which the results from a study can be generalized to a wider population.

4) When two instruments produce the same result, they are probably measuring the same construct or trait; thus they are likely to be **convergent**.

5) If an instrument has good **divergent validity**, it will be able to accurately discriminate between overlapping constructs. For example, Hamilton Anxiety Scale will be said to have high divergent validity, if it rates symptoms seen in a patient with Generalised Anxiety Disorder consistently higher than the symptoms seen in a patient with depression and some features of anxiety.

6) When testing **predictive validity**, we assess an instrument's ability to predict something it should theoretically be able to predict. In this example, the authors expect dissociation to predict later PTSD.

Table 10.5 (Continued)

Safety in epilepsy	Drug	Comments
Antipsychotics		
Low risk – good choices	Amisulpride/sulpiride	Considered to be safe in PWE. ⁴³ Renally excreted, so low risk of pharmacokinetic interactions with antiseizure medications. Seizures uncommon in overdose ⁴⁴
	Aripiprazole	Rarely lowers seizure threshold. ⁵ Incidence of seizures similar to placebo in RCTs ²⁴
	Ziprasidone	
	High potency FGAs	For example, fluphenazine, haloperidol, trifluoperazine, flupentixol. Low risk of lowering the seizure threshold ⁵
	Risperidone	Unlikely to lower the seizure threshold. ⁵ Incidence of seizures similar to placebo in RCTs. ²⁴ Has been recommended for PWE. ^{32,45} Evidence of safety in a case series of adolescents with epilepsy ⁴⁶
Probably low risk – use with caution (limited evidence)	Asenapine	Seizure rate similar to placebo in RCTs. ⁴⁷ Data and clinical experience of use in PWE is extremely limited
	Brexpiprazole	
	Cariprazine	
	Lurasidone	
Moderate risk – care required	Olanzapine	Olanzapine and quetiapine both associated with seizures in RCTs. ²⁴ However, olanzapine causes more EEG abnormalities. ⁴⁴ Overall risk of reducing the seizure threshold is considered to be low ⁵ and olanzapine has been recommended by some for PWE. ³² Data relating to olanzapine are difficult to interpret. EEG changes are seen in some but not all studies ⁴⁸ and it has been reported to be both anticonvulsant ⁴⁹ and proconvulsant. ⁵⁰ Quetiapine has a high risk of drug interaction in PWE ⁴⁵
	Quetiapine	
Higher risk – care required	Clozapine	Most epileptogenic antipsychotic. ³² However, has been used successfully in PWE stable on antiseizure medications without worsening seizures ⁵¹ and even in treatment-resistant epilepsy. ⁵² Note, should not be used with carbamazepine (risk of blood dyscrasias and reduced clozapine levels). Valproate or lamotrigine are the antiseizure medications of choice
Higher risk – avoid	Low potency FGAs (e.g. chlorpromazine)	Best avoided in PWE. ³¹ Doses of chlorpromazine above 1g/day have a 9% incidence of seizures
	Loxapine	Highest rate of seizures amongst the FGAs ⁵³
	Depot antipsychotics	None of the depot preparations currently available are thought to be epileptogenic, however: ■ The kinetics of depots are complex (seizures may be delayed) ■ If seizures do occur, the offending drug may not be easily withdrawn. Depots should be used with extreme care
	Zotepine	Has established dose-related pro-convulsive effect ⁴⁴

(Continued)